

HyperWolf: Efficient Polynomial Commitment Schemes from Lattices

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Abstract. This work introduces HyperWolf, a Hypercube-Wise Optimized polynomial commitment scheme based on Lattices over ring Fields. The scheme achieves succinctness with $O(\log N)$ proof size and verifier time, along with linear prover cost. It supports both univariate and multilinear polynomials under a unified framework. Inspired by the two-dimensional tensor structure employed in [GLS⁺21] to achieve sublinear efficiency, we generalize the idea to a k -dimensional tensor (hypercube) structure and design a k -round recursive proof protocol, where each round performs a dimensionality reduction, which results in an overall efficiency of $O(kN^{1/k})$. By setting $k = \log N$, our scheme achieves near-optimal asymptotic performance.

HyperWolf is fully transparent and relies only on the standard lattice assumption over rings. In terms of concrete efficiency, for polynomials with $N = 2^{25}$ coefficients, our scheme yields proof sizes that are $8\times$ smaller than the lattice-based scheme of [CMNW24] (Crypto24), and over $200\times$ smaller than SLAP [AFLN24] (Eurocrypt24). Compared to Greyhound [NS24] (Crypto24), our proof size is of similar magnitude, while achieving significantly lower verifier time.

Keywords: Lattice-based cryptography, Polynomial Commitment Scheme

1 Introduction

A *Polynomial Commitment Scheme (PCS)* is a cryptographic primitive that enables a prover to commit to a polynomial f —which may be univariate [KZG10] or multilinear [GLS⁺21], [ZCF24]—by publishing a compact commitment. Later, for given public inputs x (or $\vec{x}$ in the multilinear case) and y , the prover can generate a succinct proof π attesting that $y = f(x)$ or $f(\vec{x})$. For practical efficiency, the proof size must be succinct and the verification cost must be efficient (i.e. less than sublinear in the input size). PCS are widely used in cryptographic protocols, enabling applications such as verifiable secret sharing

^{**} In the future, this scheme will be implemented by Mr. Kurt Pan, and he will be included in the author list at that time.

[BDK13], secure multi-party computation [BHV⁺23], and authentication mechanisms. Their most prominent application lies in constructing zero-knowledge succinct non-interactive arguments of knowledge (zkSNARKs) [BFS20], [CHM⁺20], [CBBZ23], [BSBHR18b], where they serve as a fundamental primitive.

Significant research efforts have been made on PCS, however, the majority of existing schemes depend on discrete logarithm problem assumptions [BBB⁺18], [KZG10], [Lee21], which are vulnerable to quantum computing attacks. Although post-quantum secure alternatives have been developed - notably those employing error-correcting codes [BSBHR18a], [GLS⁺21] - these solutions typically compromise on homomorphic properties, which are essential for supporting efficient proof composition in advanced cryptographic protocols.

Lattice-based cryptography has emerged as a particularly promising approach for constructing quantum-resistant primitives, owing to its computational efficiency, natural homomorphic capabilities, and proven security against quantum attacks. Despite these advantages, the development of lattice-based polynomial commitment schemes has progressed slowly, mainly due to fundamental difficulties in achieving exact extractability guarantees. While recent work has begun to address this gap, existing solutions inevitably introduce substantial compromises. Some proposals depend on non-standard lattice assumptions [WW23a], [AFLN24], whose long-term security against quantum adversaries remains uncertain. Others require trusted setup procedures [HSS24], undermining their practical applicability. Although certain constructions [NS24] and [CMNW24] maintain transparent setup properties, they are respectively constrained by their restriction to univariate polynomials and the computational overhead associated with integer field operations. These limitations collectively underscore the importance of continued research to advance lattice-based polynomial commitment schemes.

1.1 Related Works

According to the underlying construction frameworks, the existing lattice-based polynomial commitment schemes can be broadly classified into two categories:

Constructions Based on Inner-Product Relations. These schemes interpret the polynomial evaluation process $f(x) = y$ as an inner-product relation $\langle \vec{f}, \vec{x} \rangle$, where $\vec{x} = (1, x, \dots, x^{N-1})$ in the univariate case [ACL⁺22, dCP23, WW23a, CLM23]. They leverage the Bulletproofs protocol [BBB⁺18] to generate succinct proofs. However, the verification cost is linear to the polynomial degree N because the evaluation point cannot be chosen in advance to ensure soundness. Recent works [BCS23, FLV23] address this issue by integrating the Dory-based delegation protocol [Lee21] or introduced more structured commitment scheme for fast verification, though they still necessitate a preprocessing phase. Furthermore, most constructions do not guarantee extractability. Only a few schemes [FLV23, ACL⁺22, CLM23] achieve this property, but they depend

on non-standard lattice assumptions, such as the k -M-SIS assumption, which is susceptible to being broken.¹

Constructions Based on Recursive Approaches. Another approach to constructing lattice-based polynomial evaluation schemes utilizes the recursive computational framework. [FMN24] was the first to introduce the split-and-fold technique from [BSBHR18a] into the lattice setting, enabling fast verification. This advancement, however, came at the cost of relying on the new PowerBASIS assumption, a variant of the BASIS assumption proposed in [WW23b], and required a large common reference string (CRS) of size quadratic in the input size. A subsequent work, SLAP [AFLN24], addressed these limitations by reducing the size of the CRS and mitigating the reliance on non-standard assumptions. While this refinement improved efficiency, it introduced a quasi-linear prover cost and still required a trusted setup.

Most recently, Nguyen et al. [NS24] introduced **Greyhound**, a transparent univariate polynomial commitment scheme building upon the **Labrador** protocol [BS23]. Their construction employs recursive techniques to achieve compact proof sizes while incorporating bi-variate sum-check technique to enable sub-linear verification time under standard lattice assumptions. Nevertheless, this approach maintains two key limitations: it only supports univariate polynomials and specifically operates under infinity norm bounds. Concurrently, Cini et al. [CMNW24] presented an SIS-based polynomial commitment scheme with a transparent setup that accommodates both univariate and multilinear polynomials. While this construction achieves broader functionality, it incurs substantial computational overhead due to its reliance on integer field arithmetic.

Building on the discussion above, this work explores the following research question:

Can we design a general PCS for univariate and multilinear polynomials, based on standard lattice assumptions, that supports a transparent setup, nearly optimal (logarithmic level) proof size and verification time, and linear prover time?

1.2 Our Contributions

We answer the above question in the affirmative by introducing **HyperWolf**, a novel and efficient lattice-based proof system that achieves all the desired properties. We present three main contributions, each of independent interest.

Unified Framework. First, we identify and formalize a common algebraic structure shared by univariate and multilinear polynomial evaluations, and propose a unified framework that accommodates both cases. This framework is flexibly parameterized, allowing it to specialize to either setting through the adjustment of auxiliary parameters.

Recursive-Friendly Proof System. Our second contribution is the design of a proof system works in recursive manner for the unified framework. This

¹ The k -M-SIS assumption has been demonstrated to be insecure in [WW23a].

construction is inspired by the “bivariate” paradigm, which has been widely employed in prior works such as [NS24], [GLS⁺21], and [EG25] to achieve sublinear proof sizes and verifier time. The core idea behind these schemes is to reinterpret the witness—specifically, the coefficient vector $\vec{f}$ in a PCS—as a vector of length N , and express the proof relation as an inner product $\langle \vec{f}, \vec{x} \rangle = y$. By reshaping $\vec{f}$ into an $n \times n$ matrix with $n^2 = N$, the original relation can be decomposed to two sub-relations: first, an inner product between a length- n vector and an $n \times n$ matrix; second, an inner product between two vectors of length n , which can then be proved using a bivariate sumcheck protocol. Each sub-proof yields an $O(n)$ -size proof and supports $O(n)$ -time verification, resulting in an overall sublinear proof system that maintains completeness and soundness.

In our proposed scheme, we generalize and extend this idea by reshaping a length- N witness into a k -dimensional hypercube structure, and designing a k -round recursive proof system where each round performs a reduction of knowledge from a high-dimensional relation to one with dimension decreased by one. Crucially, both the proof size and verification time in each round are bounded by $O(N^{1/k})$. After completing all k rounds, the total cost accumulates to $O(k \cdot N^{1/k})$. Notably, by optimizing the parameter choice to $k = \log N$, both the proof size and verifier time can be reduced to $O(\log N)$, achieving near-optimal asymptotic efficiency. Furthermore, our scheme maintains linear prover complexity. A comprehensive performance comparison between our scheme and existing post-quantum schemes is presented in Table 1 and Table 2.

Scheme	Assump.	Transp. Set	CRS Size	Commit. Size	Proof Size	Prover Cost	Verifier Cost	Sound. Error
[ZCF24]	Hash	Y	/	$O(1)$	$O(\log N)$	$O(N \log N)$	$O(\log^2 N)$	$\text{neg}(\lambda)$
[GLS ⁺ 21]	Hash	Y	/	$O(1)$	$O(\sqrt{N})$	$O(N)$	$O(\sqrt{N})$	$\text{neg}(\lambda)$
[FMN24]	P-BASIS	N	$O(N^2)$	$O(1)$	$O(\log N)$	$O(N^2)$	$O(\log N)$	$1/\text{poly}(\lambda)$
[AFLN24]	M-SIS(ℓ_2)	N	$O(\log N)$	$O(1)$	$O(\log^2 N)$	$O(N)$	$O(\log^2 N)$	$\text{neg}(\lambda)$
[CMNW24]	M-SIS(ℓ_∞)	Y	/	$O(\log N)$	$O(\log^2 N)$	$O(N)$	$O(\log N)$	$\text{neg}(\lambda)$
[NS24]	M-SIS(ℓ_∞)	Y	/	$O(1)$	$O(\log \log N)$	$O(N)$	$O(\sqrt{N})$	$\text{neg}(\lambda)$
HyperWolf	M-SIS(ℓ_2)	Y	/	$O(1)$	$O(\log N)$	$O(N)$	$O(\log N)$	$\text{neg}(\lambda)$

Table 1: Asymptotic performance comparison of HyperWolf and prior post-quantum PCS schemes.

Scheme	$N = 2^{15}$	$N = 2^{20}$	$N = 2^{25}$	$N = 2^{30}$
[FMN24]	–	36.5 MB	–	767 MB
[NS24]	–	–	46 KB	53 KB
[CMNW24]	120 KB	501 KB	1.51 MB	5.17 MB
HyperWolf	80.28 KB	130.44 KB	180.59 KB	230.8 KB

Table 2: Concrete efficiency comparison of lattice-based polynomial commitment schemes under 128-bit security. The table shows proof sizes (in KB/MB) for varying degrees N . Missing entries (–) indicate cases not reported in the original works.

Optimization over Ring Field. Our final contribution is to adapt the proposed proof system to operate over ring fields, which further reduces computational overhead while significantly enhancing soundness. In particular, when challenges are sampled from a ring, the challenge space becomes substantially larger under the same ℓ_2 -norm constraint. For instance, consider the ring $\mathbb{R}_q[X] := \mathbb{Z}_q[X]/(X^{64} + 1)$. Even when restricting each coefficient of a challenge polynomial within $\{0, \pm 1, \pm 2\}$, we can, by carefully controlling the frequency of each value within a given norm bound, construct a challenge space of more than 2^{128} size. See Section 5.2 for concrete details.

2 Preliminaries

2.1 Notation

Let λ be the security parameter, implicitly given to all algorithms. A function $\text{neg}(\lambda)$ is negligible, and all key parameters are assumed to be $\text{poly}(\lambda)$. For $n \in \mathbb{N}$, let $[n] := \{0, 1, \dots, n-1\}$, and define $[a:b] := \{a, a+1, \dots, b-1\}$ for integers $a < b$. Let q be an odd modulus, and define $\mathbb{Z}_q := \mathbb{Z}/q\mathbb{Z}$. Vectors in $\mathbb{Z}_q^n$ are written as $\vec{f} = (f_0, \dots, f_{n-1})$, and matrices over $\mathbb{Z}_q$ use uppercase letters (e.g., A, B). Let $\vec{1}_m$ denote the vector of length m with all entries equal to 1.

We denote by $\mathbb{Z}_q[X]$ the ring of polynomials over $\mathbb{Z}_q$. For d a power of two, define $\mathbb{R} := \mathbb{Z}[X]/(X^d + 1)$ and $\mathbb{R}_q := \mathbb{Z}_q[X]/(X^d + 1)$. Bold symbols (e.g., $\mathbf{f} \in \mathbb{R}_q$) denote ring elements, except for challenges; the constant term of $\mathbf{f}$ is $\text{ct}(\mathbf{f}) := f_0$. Bold uppercase letters denote matrices over $\mathbb{R}_q$. The concatenation of length- n vectors $\vec{\mathbf{f}}$ and $\vec{\mathbf{g}}$ is denoted by $(\vec{\mathbf{f}}, \vec{\mathbf{g}}) \in \mathbb{R}_q^{2n}$. Similarly, the concatenation of a sequence of length- n vectors $(\vec{\mathbf{f}}_i)_{i \in [k]}$ lies in $\mathbb{R}_q^{kn}$. And for matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}_q^{m \times n}$, we write $[\mathbf{A} \ \mathbf{B}] \in \mathbb{R}_q^{m \times 2n}$. For a vector $\vec{\mathbf{a}} \in \mathbb{R}_q^n$, the i -th entry is denoted $\mathbf{a}_i$ or $\mathbf{a}[i]$. Note that though $\mathbb{Z}_q$ elements can be considered as $\mathbb{R}_q$ elements with higher-degree coefficients equal to zero, we do not use boldface letters to represent them.

Norms. We define the ℓ_1 -norm, ℓ_2 -norm and ℓ_∞ -norm for $\mathbf{f} := \sum_{i=0}^{d-1} f_i X^i \in \mathbb{R}_q$ as: $\|\mathbf{f}\|_1 := \sum_{i=0}^{d-1} |f_i|$, $\|\mathbf{f}\|_2 := \sqrt{\sum_{i=0}^{d-1} f_i^2}$ and $\|\mathbf{f}\|_\infty := \max_{i \in [d]} |f_i|$. And for $\vec{\mathbf{f}} := (\mathbf{f}_0, \dots, \mathbf{f}_{m-1}) \in \mathbb{R}_q^m$, its ℓ_1 -norm, ℓ_2 -norm and ℓ_∞ -norm are:

$$\|\vec{\mathbf{f}}\|_1 := \sum_{i=0}^{m-1} \|\mathbf{f}_i\|_1, \quad \|\vec{\mathbf{f}}\|_2 := \sqrt{\sum_{i=0}^{m-1} \|\mathbf{f}_i\|_2^2}, \quad \|\vec{\mathbf{f}}\|_\infty := \max_{i \in [m]} (\|\mathbf{f}_i\|_\infty).$$

Unless otherwise specified, $\|\cdot\|$ refers to the ℓ_2 norm in this paper.

Challenge Space. Throughout the paper, we define the subset $\mathcal{C} \subset \mathbb{R}_q$ as a challenge space if the difference of any two distinct elements in $\mathcal{C}$ is invertible in $\mathbb{R}_q$. Additionally, we require that $\|c\| \leq \tau$ and $\|c\|_{\text{op}} \leq T$ for all $c \in \mathcal{C}$, where τ and T are constants in $\mathbb{Z}$. Here, $\|c\|_{\text{op}} := \sup_{\mathbf{v} \in \mathbb{R}_q} \frac{\|c\mathbf{v}\|}{\|\mathbf{v}\|}$ can be interpreted as the maximum norm of the operator c acting on any $\mathbf{v} \in \mathbb{R}_q$.

The following lemma establishes the conditions for invertibility in polynomial rings, as proposed by Lyubashevsky and Seiler [LS18].

Lemma 1. (*Invertibility Condition*) *When the prime q satisfies $q \equiv 5 \pmod{8}$, any $\mathbf{f} \in \mathbb{R}_q$ that meets the condition $0 < \|\mathbf{f}\| < q^{1/2}$ has an inverse in $\mathbb{R}_q$.*

Compose and Decompose Gadget Matrix. Let $a \in \mathbb{N}$. We define the gadget vector $\vec{g}_a := (1, a, a^2, \dots, a^{\iota-1})$, where $\iota = \lceil \log_a q \rceil$. For a matrix $\mathbf{A} \in \mathbb{R}_q^{m \times n}$, we can decompose it into $\tilde{\mathbf{A}} \in \mathbb{R}_q^{\iota m \times n}$ such that $\mathbf{A} = G_{a,m} \tilde{\mathbf{A}}$, where the gadget matrix is defined as $G_{a,m} := I_m \otimes \vec{g}_a^\top \in \mathbb{Z}_q^{m \times \iota m}$, with I_m denoting the identity matrix of size $m \times m$, and $\otimes$ denoting the Kronecker product. Formally, we define the decomposition function $G_{a,m}^{-1} : \mathbb{R}_q^{m \times n} \rightarrow \mathbb{R}_q^{\iota m \times n}$, which maps $\mathbf{A}$ to $\tilde{\mathbf{A}}$. Obviously, we have $G_{a,m} G_{a,m}^{-1}(\mathbf{A}) = \mathbf{A}$, and for each column $\tilde{\mathbf{A}}_i$ of $\tilde{\mathbf{A}}$, we have $\|\tilde{\mathbf{A}}_i\| \leq \lfloor \frac{a}{2} \rfloor \sqrt{\iota m}$ for $i \in [n]$.

The Conjugation Automorphism. Define the conjugation automorphism $\sigma_{-1} : \mathbb{R}_q \rightarrow \mathbb{R}_q$. For any $\mathbf{f} = \sum_{i=0}^{d-1} f_i X^i \in \mathbb{R}_q$, the automorphism is given by: $\sigma_{-1}(\mathbf{f}) := \sum_{i=0}^{d-1} f_i X^{-i}$. For vectors $\vec{\mathbf{a}}, \vec{\mathbf{b}} \in \mathbb{R}_q^n$, let their coefficients representations be $\vec{a}, \vec{b} \in \mathbb{Z}_q^{nd}$, respectively. The inner product between $\vec{a}$ and $\vec{b}$ is equal to the constant term of $\langle \sigma_{-1}(\vec{\mathbf{a}}), \vec{\mathbf{b}} \rangle$.

Integer-to-Ring Mapping. Given a vector $\vec{f} = (f_0, f_1, \dots, f_{nd-1}) \in \mathbb{Z}_q^{nd}$, we define the mapping $M_R : \mathbb{Z}_q^{nd} \rightarrow \mathbb{R}_q^n$ such that

$$M_R(\vec{f}) = \left(\sum_{j=0}^{d-1} f_j X^j, \sum_{j=0}^{d-1} f_{d+j} X^j, \dots, \sum_{j=0}^{d-1} f_{(n-1)d+j} X^j \right),$$

where each component is interpreted as an element in $\mathbb{R}_q$.

2.2 Lattice Problems and Ajtai Commitment

We recall the standard Module-SIS [LS15] assumption.

Definition 1. ($\text{MSIS}_{\kappa, n, q, \beta}$) *Given a random matrix $\mathbf{A} \in \mathbb{R}_q^{\kappa \times n}$ for $\kappa = \text{poly}(n)$, the goal of the problem is to find a non-zero $\vec{\mathbf{z}} \in \mathbb{R}_q^n$ such that $\mathbf{A}\vec{\mathbf{z}} = \vec{\mathbf{0}}$ over $\mathbb{R}_q$ and $\|\vec{\mathbf{z}}\| \leq \beta$.*

Micciancio and Regev [MR09] demonstrated that an MSIS algorithm is expected to produce a solution with norm $\|\vec{\mathbf{z}}\| \geq \min \left(q, 2^{2\sqrt{d\kappa} \log q \log(1.0045)} \right)$, where d is the degree of the irreducible polynomial used to construct the polynomial ring $\mathbb{R}_q$.

Ajtai commitment scheme. We review the Ajtai commitment scheme [Ajt96] where the messages are ring elements with *small* norms.

Definition 2. (*Ajtai Commitment* [Ajt96]) *Let λ be the security parameter and κ, q, β be positive integers. Suppose a prover commits to a n -dimensional short vector $\vec{\mathbf{f}} \in \mathbb{R}_q^n$ (with small norms). The Ajtai commitment is as follows:*

- $\text{Setup}(1^\lambda)$: Sample a matrix $\mathbf{G} \xleftarrow{\$} \mathbb{R}_q^{\kappa \times n}$. Output $\text{pp} := \mathbf{G}$.
- $\text{Commit}(\text{pp}, \vec{\mathbf{f}})$: Given $\vec{\mathbf{f}}$ such that $\|\vec{\mathbf{f}}\| \leq \beta$, output $\text{cm} := \mathbf{G}\vec{\mathbf{f}}$.

A commitment scheme is said to be *binding* if it is computationally infeasible to find two distinct openings for the same commitment, and *hiding* if it reveals no information about the underlying message.

The commitment cm satisfies the binding property for inputs $\|\vec{\mathbf{f}}\| < \beta$ under the assumption that $\text{MSIS}_{\kappa, n, q, 2\beta}$ is hard. Hiding can be achieved by appending a small random vector to the message. For simplicity, we treat this randomness as part of the witness throughout our protocols.

2.3 Interactive Proofs and Polynomial Commitment Schemes

Let $\mathcal{R} \subseteq \{0, 1\}^* \times \{0, 1\}^* \times \{0, 1\}^*$ be a ternary relation. For $(\text{pp}, x, w) \in \mathcal{R}$, we denote that pp as the public parameters, x as a statement and w as a witness for x . For fixed pp and x , we define the set $\mathcal{R}(\text{pp}, x) := \{w : (\text{pp}, x, w) \in \mathcal{R}\}$.

Definition 3. (Interactive Proof System) An interactive proof system $\Pi = (\mathcal{S}, \mathcal{P}, \mathcal{V})$ for the relation $\mathcal{R}$ is an interactive protocol that consists of the probabilistic polynomial-time (PPT) setup algorithm $\mathcal{S}$ and a protocol between an interactive PPT prover $\mathcal{P}$ and an interactive verifier $\mathcal{V}$. This system enables the prover to convince the verifier of the existence of a witness w for a public statement x . Both $\mathcal{P}$ and $\mathcal{V}$ receive the statement x and the public parameters pp generated by the setup algorithm $\mathcal{S}$ as public input. Additionally, $\mathcal{P}$ has private input in the form of a witness $w \in \mathcal{R}(\text{pp}, x)$. At the end of the protocol, $\mathcal{V}$ either accepts or rejects the prover's claim of knowing a witness w such that $(\text{pp}, x, w) \in \mathcal{R}$. Consequently, we classify the corresponding transcript (i.e., the collection of all messages exchanged during the protocol execution) as either accepting or rejecting.

Definition 4. (Completeness) A proof system $\Pi = (\mathcal{S}, \mathcal{P}, \mathcal{V})$ for the relation $\mathcal{R}$ satisfies completeness with completeness error $\epsilon(\cdot)$ if for all adversaries $\mathcal{A}$,

$$\Pr \left[b = 0 \wedge (\text{pp}, x, w) \in \mathcal{R} \mid \begin{array}{l} \text{pp} \leftarrow \mathcal{S}(1^\lambda) \\ (x, w) \leftarrow \mathcal{A}(\text{pp}) \\ (tr, b) \leftarrow \langle \mathcal{P}(\text{pp}, x, w), \mathcal{V}(\text{pp}, x) \rangle \end{array} \right] = \epsilon(\lambda) + \text{neg}(\lambda).$$

Furthermore, we say that Π satisfies perfect completeness when $\epsilon = 0$.

Definition 5. (Knowledge Soundness) A proof system $\Pi = (\mathcal{S}, \mathcal{P}, \mathcal{V})$ for the relation $\mathcal{R}$ is knowledge sound with knowledge error $\epsilon(\cdot)$ if there exists an expected PPT extractor $\mathcal{E}$ such that for any stateful PPT adversary $\mathcal{P}^*$:

$$\Pr \left[b = 1 \wedge (\text{pp}, x, w) \notin \mathcal{R} \mid \begin{array}{l} \text{pp} \leftarrow \mathcal{S}(1^\lambda) \\ (x, st) \leftarrow \mathcal{P}^*(\text{pp}) \\ (tr, b) \leftarrow \langle \mathcal{P}^*(\text{pp}, x, w), \mathcal{V}(\text{pp}, x) \rangle \\ w \leftarrow \mathcal{E}_{\mathcal{P}^*}(\text{pp}, x) \end{array} \right] = \epsilon(\lambda) + \text{neg}(\lambda).$$

The extractor $\mathcal{E}$ has a black-box oracle access to the (malicious) prover $\mathcal{P}^*$ and can rewind it to any point in the interaction.

We then revisit the definition of polynomial commitment schemes.

Definition 6. (Polynomial Commitment Scheme) Let $\text{PC} = (\text{Setup}, \text{Commit}, \text{Open}, \text{Eval})$ be a tuple of probabilistic polynomial-time algorithms. PC is a commitment scheme over $\mathcal{M} = \mathbb{Z}_q^{<N}[X]$ with slack space $\mathcal{SL}$ if it follows the syntax outlined below:

- $\text{Setup}(1^\lambda) \rightarrow \text{pp}$: Take security parameter λ and output public parameter pp .
- $\text{Commit}(\text{pp}, f) \rightarrow (\text{cm}, \text{st})$: Take public parameters pp , a message f and output a commitment cm and decommitment state st .
- $\text{Open}(\text{pp}, \text{cm}, f, \text{st}, s) \rightarrow 0/1$: Take public parameters pp , a commitment cm , a message f , a decommitment state st and a relaxation factor $s \in \mathcal{SL}$ and output a bit indicating whether cm is a valid commitment to f under pp .²
- $\text{Eval}(\text{pp}, \text{cm}, x, y; (f, \text{st}))$ is a protocol between the prover $\mathcal{P}$ and $\mathcal{V}$ with public parameters, commitment cm , an evaluation point x and a value y . $\mathcal{P}$ additionally knows a polynomial f and $\mathcal{P}$ wants to convince $\mathcal{V}$ that f is an opening to cm and $f(x) = y$.

The polynomial commitment scheme must satisfy three main properties: evaluation completeness, weak binding and knowledge soundness.

Definition 7. (Evaluation Completeness) A polynomial commitment scheme $\text{PC} = (\text{Setup}, \text{Commit}, \text{Open}, \text{Eval})$ satisfies completeness if for every polynomial f and any evaluation point $u \in \mathbb{Z}$:

$$\Pr \left[\begin{array}{l} \text{Eval.}\mathcal{V}(\text{pp}, \text{cm}, \\ u, f(u), \pi) = 0 \end{array} \middle| \begin{array}{l} \text{pp} \leftarrow \mathcal{S}(1^\lambda) \\ (\text{cm}, \text{st}) \leftarrow \text{Commit}(\text{pp}, f) \\ \pi \leftarrow \text{Eval.}\mathcal{P}(\text{pp}, \text{cm}, u, f(u); (f, \text{st})) \end{array} \right] = \epsilon(\lambda) + \text{neg}(\lambda).$$

Definition 8. (Weak Binding) A polynomial commitment scheme $\text{PC} = (\text{Setup}, \text{Commit}, \text{Open}, \text{Eval})$ satisfies weak binding if for every PPT adversary $\mathcal{A}$:

$$\Pr \left[\begin{array}{l} f \neq f' \wedge f, f' \in R^{<N}[X] \wedge \\ \text{Open}(\text{pp}, \text{cm}, f, \text{st}, c) \\ = \text{Open}(\text{pp}, \text{cm}, f', \text{st}', c') = 1 \end{array} \middle| \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda) \\ (\text{cm}, (f, \text{st}, c), (f', \text{st}', c')) \leftarrow \mathcal{A}(\text{pp}) \end{array} \right] = \text{neg}(\lambda).$$

Definition 9. (Knowledge Soundness) A polynomial commitment scheme $\text{PC} = (\text{Setup}, \text{Commit}, \text{Open}, \text{Eval})$ is knowledge sound with knowledge error $\epsilon(\cdot)$ if for all stateful PPT adversary $\mathcal{P}^*$, there exists an expected PPT extractor $\mathcal{E}$ such that:

$$\Pr \left[\begin{array}{l} o = \text{Open}(\text{pp}, \text{cm}, f, \text{st}, s) \\ b = 1 \wedge (o \neq 1 \vee f(u) \neq z) \end{array} \middle| \begin{array}{l} \text{pp} \leftarrow \mathcal{S}(1^\lambda) \\ (\text{cm}, u, z, \pi) \leftarrow \mathcal{P}^*(\text{pp}) \\ b \leftarrow \text{Eval.}\mathcal{V}(\text{pp}, \text{cm}, u, z, \pi) \\ (f, \text{st}, s) \leftarrow \mathcal{E}_{\mathcal{P}^*}(\text{pp}, \text{cm}, u, z, \pi) \end{array} \right] = \epsilon(\lambda) + \text{neg}(\lambda).$$

² We implicitly assume $b = 0$ if $s \notin \mathcal{SL}$.

2.4 Coordinate-Wise Special Soundness

Coordinate-Wise Special Soundness is an extension of the concept of special soundness, which can lead to knowledge soundness for a $(2\mu + 1)$ -round interactive proof system, as proposed in [FMN24].

First, consider two vectors $\vec{x} := (x_1, x_2, \dots, x_\ell), \vec{y} := (y_1, y_2, \dots, y_\ell) \in S^\ell$. We define a relation $\equiv_i$ for $i \in [\ell]$ as follows:

$$\vec{x} \equiv_i \vec{y} \Leftrightarrow x_i \neq y_i \wedge \forall j \in [\ell] \setminus \{i\}, x_j = y_j.$$

This means that the vectors $\vec{x}$ and $\vec{y}$ are identical in all coordinates except for the i -th one. Next, we define the set:

$$SS(S, \ell, k) := \left\{ \{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_K\} \subseteq (S^\ell)^K \mid \begin{array}{l} \exists e \in [K], \forall i \in [\ell], \\ \exists J = \{j_1, \dots, j_{k-1}\} \subseteq [K] \setminus \{e\}, \\ \forall j \in J, \vec{x}_e \equiv_i \vec{x}_j \end{array} \right\},$$

where $K := \ell(k - 1) + 1$. The interpretation of the set $SS(S, \ell, k)$ is that there exists a “central” vector $\vec{x}_e$ such that for each coordinate of $\vec{x}_e$, there are $k - 1$ other vectors that differ from it only at that coordinate. In essence, for every coordinate, there are k vectors that differ from each other only in that coordinate with $\vec{x}_e$ always being one of them.

Definition 10. (CWSS for Multi-Round Protocols) Let $\Pi = (S, \mathcal{P}, \mathcal{V})$ be public-coin $(2\mu + 1)$ -round interactive proof system for relation $\mathcal{R}$, where in each round the challenge space of $\mathcal{V}$ is defined as $\mathcal{C} = S^\ell$. A tree of transcripts is a set of $K = (\ell(k - 1) + 1)^\mu$ arranged in the tree structure: The nodes in the tree correspond to the prover’s messages and the edges correspond to the verifier’s challenges. Each node at depth i has exactly $\ell(k - 1) + 1$ children corresponding to $\ell(k - 1) + 1$ distinct challenges which, as a set of vectors, lie in $SS(S, \ell, k)$. Every transcript corresponds to exactly one path from the root to a leaf node. We say that Π is ℓ -coordinate-wise k -special sound if there is a polynomial time algorithm that, given a statement x and the tree of transcripts, outputs a witness $w \in \mathcal{R}(i, x)$.

Notably, when $\ell = 1$, this definition reduces to the standard k -special soundness property.

Definition 11. (Round-by-Round Coordinate-Wise Special Soundness) Let $\Pi = (S, \mathcal{P}, \mathcal{V})$ be a $(2\mu + 1)$ -message interactive proof system structured in a recursive manner for a relation $\mathcal{R}$. The protocol Π is said to be round-by-round ℓ -coordinate-wise k -special sound for the sequence of relations $(\mathcal{R}_{\mu-i})_{i \in [\mu+1]}$ if the challenge space in each round is S^ℓ , and for any $i \in [\mu]$, there exists an extractor Ext_i such that, given a statement x_i and a set of $\ell(k - 1) + 1$ accepting transcripts $(\pi_i, c_{i,j}, w_{i+1,j})_{j \in [\ell(k-1)+1]}$, with the challenges $(c_{i,j})_{j \in [\ell(k-1)+1]}$ belonging to $SS(S, \ell, k)$, where π_i is the i -th round common first message, each $w_{i+1,j}$ corresponds to a next-round witness satisfying $(x_{i+1,j}, w_{i+1,j}) \in \mathcal{R}_{\mu-i-1}$, the extractor returns a witness w_i such that $(x_i, w_i) \in \mathcal{R}_{\mu-i}$.

Lemma 2. (Theorem 2 in [CMNW24]): Let $\Pi = (\mathcal{S}, \mathcal{P}, \mathcal{V})$ be a $(2\mu + 1)$ -message public-coin interactive proof system structured in a recursive manner for a relation $\mathcal{R}$ with associated relations $(\mathcal{R}_{\mu-i})_{i \in [\mu+1]}$. If Π is round-by-round ℓ -coordinate-wise k special sound for the tuple of relations $(\mathcal{R}_{\mu-i})_{i \in [\mu+1]}$ and $(\ell + 1)^\mu = \text{poly}(\lambda)$, then Π is knowledge sound for the relation $\mathcal{R}$ with knowledge error $\mu\ell(k - 1)/|S|$.

2.5 Reduction of Knowledge

We recall the definition of reduction of knowledge from [KP23].

Definition 12. (Reduction of Knowledge [KP23]) Let λ be a security parameter. Given two relations $\mathcal{R}_1$ and $\mathcal{R}_2$, denote $\langle \mathcal{P}, \mathcal{V} \rangle$ be an interactive protocol between a prover $\mathcal{P}$ and a verifier $\mathcal{V}$. A reduction of knowledge protocol Π from $\mathcal{R}_1$ to $\mathcal{R}_2$ consists of the following algorithms:

- $\text{Setup}(1^\lambda) \rightarrow \text{pp}$: Output public parameters pp .
- $\langle \mathcal{P}(\text{pp}, u_1, w_1), \mathcal{V}(\text{pp}, u_1) \rangle \rightarrow (u_2, w_2)$: On input public parameters pp , a shared public statement u_1 , $\mathcal{P}$ (with a witness w_1 such that $(u_1, w_1) \in \mathcal{R}_1$) and $\mathcal{V}$ engage in Π . At the end of Π , $\mathcal{P}$ and $\mathcal{V}$ output u_2 (or $\perp$ if abort), $\mathcal{P}$ additionally outputs w_2 .

Reduction of knowledge protocols satisfy the following properties:

Completeness. For every PPT adversary $\mathcal{A}$ that adaptively chooses an $\mathcal{R}_1$ instance $(u_1, w_1) \leftarrow \mathcal{A}(\text{pp})$ when given $\text{pp} \leftarrow \text{Setup}(1^\lambda)$, the protocol execution $(u_2, w_2) \leftarrow \langle \mathcal{P}(\text{pp}, u_1, w_1), \mathcal{V}(\text{pp}, u_1) \rangle$ satisfies $(u_2, w_2) \in \mathcal{R}_2$ if $(u_1, w_1) \in \mathcal{R}_1$.

Knowledge soundness. For every expected polynomial-time adversary $\mathcal{A}$ and malicious prover $\mathcal{P}^*$, there is an expected polynomial-time extractor $\mathcal{E}$ such that given $\text{pp} \leftarrow \text{Setup}(1^\lambda)$ and $(u_1, w_1^*) \leftarrow \mathcal{A}(\text{pp})$,

$$\Pr[(u_1, \mathcal{E}(\text{pp}, u_1, w_1^*)) \in \mathcal{R}_1] \approx \Pr[\langle \mathcal{P}^*(\text{pp}, u_1, w_1^*), \mathcal{V}(\text{pp}, u_1) \rangle \in \mathcal{R}_2].$$

Public reducibility. There is a deterministic polynomial-time algorithm f , such that for any PPT adversary $\mathcal{A}$ and a malicious expected polynomial-time prover $\mathcal{P}^*$, given

$$\text{pp} \leftarrow \text{Setup}(1^\lambda), \quad (u_1, w_1^*) \leftarrow \mathcal{A}(\text{pp}), \quad (u_2, w_2) \leftarrow \langle \mathcal{P}^*(\text{pp}, u_1, w_1^*), \mathcal{V}(\text{pp}, u_1) \rangle,$$

and the transcript tr , we have $f(\text{pp}, u_1, \text{tr}) = u_2$.

By the composition theory [KP23], the reduction of knowledge protocols can be composed. For two (or multiple) reduction of knowledge protocols Π_1 and Π_2 , they can be composed in two ways [KP23]: *sequential composition* $\Pi_1 \diamond \Pi_2$ and *parallel composition* $\Pi_1 \times \Pi_2$.

Theorem 1. Sequential composition (Theorem 5 in [KP23]). Let $\mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3$ be three relations. Given two reduction of knowledge protocols, Π_1 from $\mathcal{R}_1$ to $\mathcal{R}_2$ and Π_2 from $\mathcal{R}_2$ to $\mathcal{R}_3$, the composed protocol $\Pi_1 \diamond \Pi_2$ is a reduction of knowledge from $\mathcal{R}_1$ to $\mathcal{R}_3$.

Theorem 2. *Parallel composition (Theorem 6 in [KP23]).* Let $\mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3, \mathcal{R}_4$ be four relations. Given two reduction of knowledge protocols, Π_1 from $\mathcal{R}_1$ to $\mathcal{R}_2$ and Π_2 from $\mathcal{R}_3$ to $\mathcal{R}_4$, the composed protocol $\Pi_1 \times \Pi_2$ is a reduction of knowledge from $(\mathcal{R}_1 \times \mathcal{R}_3)$ to $(\mathcal{R}_2 \times \mathcal{R}_4)$.

2.6 Modular Johnson-Lindenstrauss Lemma

To prove that a vector $\vec{a} \in \mathbb{Z}^n$ has a small ℓ_2 -norm without revealing the entire vector, we can utilize the Johnson-Lindenstrauss (JL) Lemma. The key intuition behind the JL Lemma is that a random linear projection of the original vector approximately preserves its ℓ_2 -norm. Specifically, instead of disclosing $\vec{a}$ directly, the verifier can generate a random linear matrix $\Pi \in \mathbb{Z}^{256 \times n}$, where the entries of Π are independent and take values $-1, 0$, and 1 with probabilities $1/4, 1/2$, and $1/4$, respectively. The prover then reveals the projected vector $\vec{p} = \Pi \vec{a}$. The norm of $\vec{p}$ can then be used to approximate and infer the norm of $\vec{a}$ through the following lemma [GHL22], [BS23].

Lemma 3. *Modular Johnson-Lindenstrauss Variant:* Let $q \in \mathbb{N}$, $\mathcal{D}$ be a distribution on $\{0, \pm 1\}$ such that $\mathcal{D}(1) = \mathcal{D}(-1) = \frac{1}{4}$ and $\mathcal{D}(0) = \frac{1}{2}$. For every vector $\vec{a} \in \mathbb{Z}_q^n$ with $\|\vec{a}\| \leq b$ and $b \leq q/125$, we have:

$$\Pr_{\Pi \leftarrow \mathcal{D}^{256 \times n}} [\|\Pi \vec{a} \bmod q\|^2 < 30b^2] \gtrsim 2^{-128}.$$

3 Technique Overview

In this section, we present the core intuition and techniques of our work. Our goal is to construct a unified polynomial commitment scheme that supports both univariate and multilinear polynomials. To accomplish this, we first analyze their evaluation procedures, extract a common structural pattern, and abstract them into a general relation. We then design a proof system for this relation, which, by instantiating appropriate parameters, yields concrete polynomial commitment schemes for both cases.

3.1 Analysis of the Polynomial Evaluation Procedure

We begin by reviewing the evaluation procedures for univariate and multilinear polynomials. As shown in [GLS⁺21], both can be expressed uniformly as:

$$y = \left(\begin{bmatrix} f_0 & f_1 & \cdots & f_{b_0-1} \\ f_{b_0} & f_{b_0+1} & \cdots & f_{2b_0-1} \\ \vdots & \vdots & \ddots & \vdots \\ f_{(b_1-1)b_0} & f_{(b_1-1)b_0+1} & \cdots & f_{b_1b_0-1} \end{bmatrix} \cdot \vec{a}_0 \right) \cdot \vec{a}_1, \quad (1)$$

where $N = b_1 b_0$ is the total number of coefficients, and $\vec{a}_0, \vec{a}_1$ are auxiliary evaluation vectors.

In the univariate setting, the polynomial is given by $f(X) := \sum_{i=0}^{N-1} f_i X^i$ and the evaluation point is $u \in \mathbb{Z}_q$, we define the vectors $\vec{a}_0$ and $\vec{a}_1$ as follows:

$$\begin{aligned}\vec{a}_0 &:= \bigotimes_{i=0}^{\log b_0 - 1} (1, u^{2^i}) = (1, u, \dots, u^{b_0-1}) \in \mathbb{Z}_q^{b_0}, \\ \vec{a}_1 &:= \bigotimes_{i=0}^{\log b_1 - 1} (1, u^{2^i b_0}) = (1, u^{b_0}, \dots, u^{(b_1-1)b_0}) \in \mathbb{Z}_q^{b_1}.\end{aligned}$$

While in the multilinear setting, the polynomial is defined as $f(X_0, X_1, \dots, X_{\ell-1}) := f_0 + f_1 X_0 + f_2 X_1 + \dots + f_{N-1} X_0 X_1 \dots X_{\ell-1}$ for $N = 2^\ell$ and the evaluation point is $\vec{u} \in \mathbb{Z}_q^\ell$, the vectors $\vec{a}_0$ and $\vec{a}_1$ are :

$$\vec{a}_0 := \bigotimes_{i=0}^{\log b_0 - 1} (1, u_i) \in \mathbb{Z}_q^{b_0}, \quad \vec{a}_1 := \bigotimes_{i=\log b_0}^{\ell-1} (1, u_i) \in \mathbb{Z}_q^{b_1}.$$

Here $\otimes$ denotes the tensor product operator, which essentially corresponds to the Kronecker product applied sequentially to multiple vectors.

It can be observed that $\vec{a}_1 \otimes \vec{a}_0$ corresponds to the evaluation vector $\vec{x}$, which is either $(1, u, \dots, u^{N-1})$ in the univariate case or $(1, u_0, u_1, \dots, u_0 u_1 \dots u_{\ell-1})$ in the multilinear case. By first folding the coefficient matrix column-wise using $\vec{a}_0$ and then aggregating the result with $\vec{a}_1$, one obtains a value equal to the evaluation result $\langle \vec{f}, \vec{x} \rangle$ while achieving sublinear proof size and sublinear verification time, as demonstrated in [GLS⁺21, NS24].

The key contribution of this paper is to further optimize this structure and generalize it to a k -dimensional setting. Specifically, we reshape the length- N coefficient vector, where $N = \prod_{i=0}^{k-1} b_i$, into a $b_{k-1} \times \dots \times b_0$ hypercube.³ Accordingly, we prepare k evaluation vectors $\vec{a}_0, \dots, \vec{a}_{k-1}$. At each step, $\vec{a}_i$ folds the b_i slices along the $i+1$ -th dimension, reducing the hypercube's dimension by one. This process continues until a one-dimensional vector remains, whose inner product with $\vec{a}_{k-1}$ yields the final evaluation. Figure 1 illustrates the case for $k = 3$. For the univariate case, $\vec{a}_i = \left(1, u^{\prod_{j=0}^{i-1} b_j}, u^{2 \prod_{j=0}^{i-1} b_j}, \dots, u^{(b_i-1) \prod_{j=0}^{i-1} b_j}\right)$ for $i \in [k]$. For the multivariate case, $\vec{a}_i = \bigotimes_{j=\sum_{l=0}^{i-1} \log b_l}^{\sum_{l=0}^i \log b_l - 1} (1, u_l)$. And in such case we have $\bigotimes_{i=0}^{k-1} \vec{a}_i = \vec{x}$. The evaluation process is defined as:

$$y = (\mathcal{F}(\dots(\mathcal{F}(\mathcal{F}([F]) \cdot \vec{a}_0) \cdot \vec{a}_1) \cdot \vec{a}_2) \dots) \cdot \vec{a}_{k-1}, \quad (2)$$

Here, $[F]$ denotes a k -dimensional coefficient hypercube of size $b_{k-1} \times b_{k-2} \times \dots \times b_0$. In the special case where $k = 2$, $[F]$ reduces to a coefficient matrix, as defined in Equation 1. The function $\mathcal{F}$ maps a $(k-i)$ -dimensional hypercube of size $b_{k-1} \times b_{k-2} \times \dots \times b_i$ to a “ $(k-i-1)$ -dimensional hypercube matrix” of size $(b_{k-1} \dots b_{i+1}) \times b_i$, for each $i \in [k]$.

³ Unlike standard MLEs, we do not assign coefficients to Boolean vertices but arrange them in a multidimensional array.

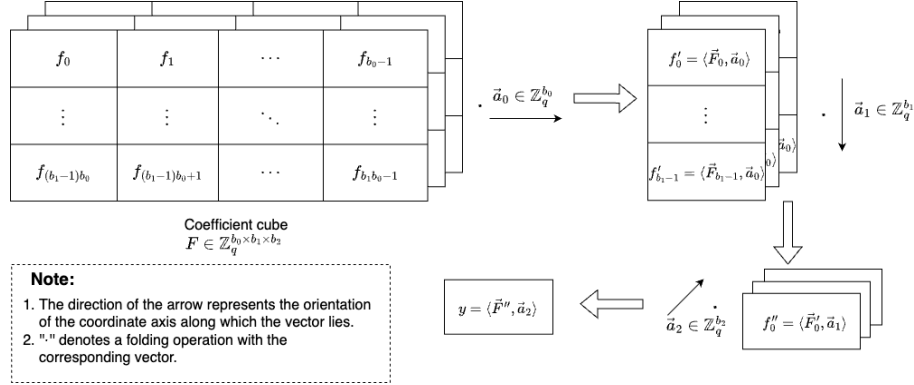


Fig. 1: Example of the evaluation process for $k = 3$.

3.2 Generalized Relation for PCS

Building on the above analysis, we abstract a unified relation that can be adapted for evaluating both univariate and multilinear polynomials.

The witness is derived from the coefficient hypercube described above. To ensure compatibility with the Ajtai commitment scheme, a decomposition procedure followed by a mapping to a ring representation should be applied to obtain a small-norm hypercube over the ring. For commitment purposes, this hypercube should be further flattened into multiple one-dimensional vectors. Besides, the witness must satisfy the relation specified in Equation 2, and the public parameters include the evaluation vectors $(\vec{a}_j)_{j \in [k]}$ defined earlier. Accounting for the ring transformation factor, we define $N = \left(\prod_{i=0}^{k-1} b_i\right) d$, and the relation is formally defined as:

$$\mathcal{R}_k = \left\{ \begin{array}{l} x = \left(\text{pp}^{(k)}, \text{cm}_{\text{out}}^{(k)}, \mathbf{y}^{(k)}, (\vec{a}_j)_{j \in [k]} \right); \\ w = \left(\mathbf{s}^{(k)}, \left(D(\mathbf{s}_i^{(k-1)}), \text{cm}_{\text{in},i}^{(k-1)} \right)_{i \in [b_{k-1}]} \right) : \\ \forall i \in [b_{k-1}], \text{Com} \left(D(\mathbf{s}_i^{(k-1)}) \right) = \text{cm}_{\text{in},i}^{(k-1)}, \|D(\mathbf{s}_i^{(k-1)})\| \leq \beta^{(k-1)}, \\ \hat{\mathbf{t}}_c = G^{-1}((\text{cm}_{\text{in},i}^{(k-1)})_{i \in [b_{k-1}]}, \text{cm}_{\text{out}}^{(k)} = \text{Com}(\hat{\mathbf{t}}_c), \\ \langle \text{Fold}^{(k)}(\mathbf{s}^{(k)}, (\vec{a}_j)_{j \in [k-1]}), \vec{a}_{k-1} \rangle = \mathbf{y}^{(k)} \end{array} \right. \quad (3)$$

Here, the operator $D : \mathbb{F}^{b_{r-1} \times b_{r-2} \times \dots \times b_0} \rightarrow \mathbb{F}^{\prod_{j=0}^{r-1} b_j}$ is defined as a flattening function, which maps an r -dimensional hypercube to a one-dimensional vector by sequentially concatenating all its "edges", for any $r \in [k]$.

The statement includes the commitment parameters $\text{pp}^{(k)} = (\mathbf{A}^{(k)}, \mathbf{B}^{(k)})$ for the relation $\mathcal{R}_k$, the outer commitment $\text{cm}_{\text{out}}^{(k)}$ (which binds with the inner

commitments $G^{-1}((\text{cm}_{\text{in},i})_{i \in [b_{k-1}]})$, a ring element $\mathbf{y}^{(k)}$ (whose constant term corresponds to the evaluation result in the evaluation scenario), and k integer vectors $(\vec{a}_j)_{j \in [k]}$ (which correspond to the auxiliary vectors as described in Section 3.1).

The witness consists of the following three components:

- A k -dimensional hypercube $\mathbf{s}^{(k)} \in \mathbb{R}_q^{b_{k-1} \times b_{k-2} \times \dots \times b_0 \iota}$, which corresponds to the coefficient hypercube described above.
- A set of b_{k-1} one-dimensional vectors, each of length $(\prod_{j=0}^{k-2} b_j) \iota$, where each vector is the flattened version of $\mathbf{s}_i^{(k-1)} \in \mathbb{R}_q^{b_{k-2} \times b_{k-3} \times \dots \times b_0 \iota}$, which is actually the i -th slice of $\mathbf{s}^{(k)}$ along the k -th dimension.
- The Ajtai commitment of each flattened vector $D(\mathbf{s}_i^{(k-1)})$.

The witness must satisfy the following three conditions:

1. Binds with the inner/ outer commitments;
2. Each flattened vector $D(\mathbf{s}_i^{(k-1)})$ must satisfy a prescribed norm bound;
3. The evaluation result $\mathbf{y}^{(k)}$ must equal the inner product between the folding of the coefficient hypercube $\text{Fold}^{(k)}(\mathbf{s}^{(k)}, (\vec{a}_j)_{j \in [k-1]})$ and the vector $\vec{a}_{k-1}$, where the fold operation is given by:

$$\begin{aligned} & \text{Fold}^{(k)}(\mathbf{s}^{(k)}, (\vec{a}_j)_{j \in [k-1]}) \\ &= \left(\mathcal{F} \left(\dots \left(\mathcal{F} \left(\mathcal{F}(\mathbf{s}^{(k)}) \cdot \sigma_{-1}(M_R(\vec{a}_0)) \right) \cdot \vec{a}_1 \right) \cdot \vec{a}_2 \right) \dots \right) \cdot \vec{a}_{k-2} \end{aligned} \quad (4)$$

3.3 Proof System for $\mathcal{R}_k$

The proof system for $\mathcal{R}_k$ aims to show that a witness satisfies the three constraints in Section 3.2. It adopts the structure of the sum-check protocol to recursively reduce the verification of a high-dimensional witness to that of lower dimensions, continuing until the base case.

To verify the norm constraint, the verifier samples $\Pi \leftarrow \mathcal{D}^{256 \times b_0 \iota}$ and sends it to the prover. The prover computes the projection $\vec{\mathbf{p}}_i^{(k)} = \sigma_{-1}(\Pi^{(k)}) \cdot D(\mathbf{s}_i^{(k-1)})$, where $\Pi^{(k)} = \vec{\mathbf{1}}^T \otimes \Pi$ with $\vec{\mathbf{1}} \in \mathbb{Z}_q^{\prod_{j=1}^{k-2} b_j}$ and sends to the verifier. The verifier then checks:

$$\sum_{j=0}^{255} \left(\text{ct}(\vec{\mathbf{p}}_{i,j}^{(k)}) \right)^2 \leq 128 \cdot (\beta^{(k-1)})^2, \quad \forall i \in [b_{k-1}],$$

which ensures the norm bound with high probability under that case $\vec{\mathbf{p}}_i^{(k)}$ is well computed according to Lemma 3.

For the inner product and binding constraints, the prover sends $\text{fold}^{(k)} = \text{Fold}^{(k)}(\mathbf{s}^{(k)}, (\vec{a}_j)_{j \in [k-1]})$ and $(\text{cm}_{\text{in},i}^{(k-1)})$ to the verifier. The verifier then checks whether $\langle \text{fold}^{(k)}, \vec{a}_{k-1} \rangle = \mathbf{y}^{(k)}$ and verifies the binding relation between $\text{cm}_{\text{out}}^{(k)}$ and $(\text{cm}_{\text{in},i}^{(k-1)})_{i \in [b_{k-1}]}$, i.e. $\mathbf{B}^{(k)}(G_{\delta_t, b_{k-1} \kappa}^{-1}((\text{cm}_{\text{in},i}^{(k-1)})_{i \in [b_{k-1}]}) \stackrel{?}{=} \text{cm}_{\text{out}}^{(k)})$.

To validate that $\text{fold}^{(k)}$ and $(\vec{\mathbf{p}}_i^{(k)}, \text{cm}_{\text{in},i}^{(k-1)})$ are honestly computed, the verifier samples a challenge $\vec{C}^{(k)} \in \mathcal{C}^{b_{k-1}}$. The prover updates the witness by using $\vec{C}^{(k)}$ to fold the b_{k-1} $(k-1)$ -dimensional components of $\mathbf{s}^{(k)}$ along the k -th dimension into a new $(k-1)$ -dimensional witness $\mathbf{s}^{(k-1)}$, computing flattened vectors $D(\mathbf{s}_i^{(k-2)})$ and corresponding commitments $\text{cm}_{\text{in},i}^{(k-2)}$. Both parties update the public parameters accordingly.

Next, the prover computes the projected vectors $\vec{\mathbf{p}}_i^{(k-1)} = \sigma_{-1}(\mathbf{\Pi}^{(k-1)})D(\mathbf{s}_i^{(k-2)})$, and sends to the verifier $\text{fold}^{(k-1)} = \text{Fold}^{(k-1)}(\mathbf{s}^{(k-1)}, (\vec{a}_j)_{j \in [k-2]})$ along with the tuple $(\vec{\mathbf{p}}_i^{(k-1)}, \text{cm}_{\text{in},i}^{(k-2)})$ for $i \in [b_{k-2}]$.

The verifier then performs the following checks:

1. $\langle \text{fold}^{(k-1)}, \vec{a}_{k-2} \rangle \stackrel{?}{=} \langle \text{fold}^{(k)}, \vec{C}^{(k)} \rangle$;
2. $\sum_{i=0}^{b_{k-1}-1} C_i^{(k)} \vec{\mathbf{p}}_i^{(k)} \stackrel{?}{=} \sum_{i=0}^{b_{k-2}-1} \vec{\mathbf{p}}_i^{(k-1)}$;
3. $\sum_{j=0}^{255} (\text{ct}(\vec{\mathbf{p}}_{i,j}^{(k-r)}))^2 \leq 128(\beta^{(k-r-1)})^2, \forall i \in [b_{k-r-1}]$;
4. Verification of the binding relation between the updated $\text{cm}_{\text{out}}^{(k-1)}$ and $(\text{cm}_{\text{in},i}^{(k-2)})_{i \in [b_{k-2}]}$.

The verifier then samples a new challenge $\vec{C}^{(k-1)} \in \mathcal{C}^{b_{k-2}}$, and the prover recursively reduces the witness dimension by repeating the folding steps. This process continues until the witness is reduced to a one-dimensional vector $\vec{\mathbf{s}}^{(1)}$, which the prover sends to the verifier for the final consistency and norm checks.

A simple example for the case $k = 3$ is shown in Appendix A.

4 HyperWolf: Efficient Proofs for the Generalized Relation

This section presents an efficient protocol for proving the relation $\mathcal{R}_k$ defined in Equation 3. We start by introducing the core proof system and then proceed with an analysis of its security and efficiency.

4.1 Core Protocol

The core protocol for proving the relation $\mathcal{R}_k$ follows the structure of the sum-check protocol and can be interpreted as a composition of $(k-1)$ reduction of knowledge protocols organized in a recursive manner. For $r \in [k-1]$, the goal of Round r is to reduce the relation $\mathcal{R}_{k-r}$ to $\mathcal{R}_{k-r-1}$, where $\mathcal{R}_{k-r}$ is defined analogously to $\mathcal{R}_k$ in Equation 3, with k replaced by $k-r$. In Round $k-1$, the prover sends the witness $\vec{\mathbf{s}}^{(1)}$ directly to the verifier, who checks whether the output of round $k-2$ satisfies the base relation $\mathcal{R}_1$. The complete protocol is described in Protocol 1.

At the start of the protocol (the initial phase), the verifier samples a random matrix $\mathbf{\Pi} \in \{-1, 0, 1\}^{256 \times b_0 \cdot d}$ from a distribution satisfying the John-Lindenstrauss (JL) properties, as stated in Lemma 3.

Protocol 1 Core Protocol for Proving $\mathcal{R}_k$

Public Inputs:

$\mathbf{A} \in \mathbb{R}_q^{\kappa \times b_0 \iota}$, $\mathbf{B} \in \mathbb{R}_q^{\kappa \times \kappa \iota'}$, $(\vec{a}_j \in \mathbb{Z}_q^{b_j})_{j \in [k] \setminus \{0\}}$, $\vec{a}_0 \in \mathbb{Z}_q^{b_0 \iota}$, $\mathbf{y}^{(k)} \in \mathbb{Z}_q$, $\mathbf{cm}_{\text{out}}^{(k)} \in \mathbb{R}_q^\kappa$

Private Inputs:

$\mathbf{s}^{(k)} \in \mathbb{R}_q^{b_{k-1} \times \dots \times b_0 \iota}$, $\left(D(\mathbf{s}_i^{(k-1)}) \in \mathbb{R}_q^{(\prod_{j=0}^{k-2} b_j) \iota}, \mathbf{cm}_{\text{in},i}^{(k-1)} \in \mathbb{R}_q^\kappa \right)_{i \in [b_{k-1}]}$

Initial Phase:

– $\mathcal{V} \rightarrow \mathcal{P} : \Pi \leftarrow \mathcal{D}^{256 \times b_0 \iota}$.

For $r = 0$ to $k - 2$, **do**: $\mathcal{R}_{k-r} \rightarrow \mathcal{R}_{k-r-1}$

- $\mathcal{P}$ and $\mathcal{V}$ construct $\mathbf{A}^{(k-r)}$, $\mathbf{B}^{(k-r)}$, $\Pi^{(k-r)}$.
- $\mathcal{P}$ computes $\vec{\mathbf{p}}_i^{(k-r)} = \sigma_{-1}(\Pi^{(k-r)}) D(\mathbf{s}_i^{(k-r-1)})$, $\text{fold}^{(k-r)}$.
- $\mathcal{P} \rightarrow \mathcal{V} : \pi_r = \left((\vec{\mathbf{p}}_i^{(k-r)})_{i \in [b_{k-r-1}]}, \text{fold}^{(k-r)}, (\mathbf{t}_i^{(k-r-1)})_{i \in [b_{k-r-1}]} \right)$.
- $\mathcal{V}$ checks:
 1. $\langle \text{fold}^{(k-r)}, \vec{a}_{k-r-1} \rangle \stackrel{?}{=} \mathbf{y}^{(k-r)}$;
 2. $\sum_{j=0}^{255} \left(\text{ct}(\vec{\mathbf{p}}_{i,j}^{(k-r)}) \right)^2 \leq 128(\beta^{(k-r-1)})^2, \quad \forall i \in [b_{k-r-1}]$;
 3. Let $\hat{\mathbf{t}}_c = G_{\delta_t, b_{k-r-1} \kappa}^{-1} \left((\mathbf{cm}_{\text{in},i}^{(k-r-1)})_{i \in [b_{k-r-1}]} \right)$,
 verify: $\mathbf{B}^{(k-r)} \hat{\mathbf{t}}_c \stackrel{?}{=} \mathbf{cm}_{\text{out}}^{(k-r)}$;
 4. If $r > 0$, additionally checks:

$$\sum_{j=0}^{b_{k-r}-1} C_j^{(k-r+1)} \vec{\mathbf{p}}_j^{(k-r+1)} \stackrel{?}{=} \sum_{j=0}^{b_{k-r-1}-1} \vec{\mathbf{p}}_j^{(k-r)}.$$

- $\mathcal{V} \rightarrow \mathcal{P} : \vec{C}^{(k-r)} \in \mathcal{C}^{b_{k-r-1}}$.
- $\mathcal{P}$ computes $w_{r+1} = \left(\mathbf{s}^{(k-r-1)}, (D(\mathbf{s}_i^{(k-r-2)}), \mathbf{cm}_{\text{in},i}^{(k-r-2)})_{i \in [b_{k-r-2}]} \right)$:
 - $\mathbf{s}^{(k-r-1)} = \sum_{i=0}^{b_{k-r-1}-1} C_i^{(k-r)} \mathbf{s}_i^{(k-r-2)}$;
 - $\mathbf{cm}_{\text{in},i}^{(k-r-2)} = \mathbf{A}^{(k-r-1)} D(\mathbf{s}_i^{(k-r-2)})$.
- $\mathcal{P}$ and $\mathcal{V}$ compute: $x_{r+1} = (\text{pp}^{(k-r-1)}, (\vec{a}_j)_{j \in [b_{k-r-2}]}, \mathbf{y}^{(k-r-1)}, \mathbf{cm}_{\text{out}}^{(k-r-1)})$:
 - $\mathbf{y}^{(k-r-1)} = \langle \text{fold}^{(k-r)}, \vec{C}^{(k-r)} \rangle$;
 - $\mathbf{cm}' = \sum_{i=0}^{b_{k-r-1}-1} C_i^{(k)} \mathbf{cm}_{\text{in},i}^{(k-r-1)}, \mathbf{cm}_{\text{out}}^{(k-r-1)} = \mathbf{B} \left(G_{\delta_t, \kappa}^{-1}(\mathbf{cm}') \right)$.

Final Check (Round $k - 1$):

- $\mathcal{P} \rightarrow \mathcal{V} : \mathbf{s}^{(1)}$.
 - $\mathcal{V}$ checks:
 1. $\langle \sigma_{-1}(M_R(\vec{a}_0)), \mathbf{s}^{(1)} \rangle \stackrel{?}{=} \langle \text{fold}^{(2)}, \vec{C}^{(2)} \rangle = \mathbf{y}^{(1)}$;
 2. $\sigma_{-1}(\Pi) \mathbf{s}^{(1)} \stackrel{?}{=} \sum_{i=0}^{b_1-1} C_i^{(2)} \vec{\mathbf{p}}_i^{(2)}$;
 3. $\|\mathbf{s}^{(1)}\| \leq \beta^{(0)}$; $\mathbf{A} \mathbf{s}^{(1)} \stackrel{?}{=} \sum_{i=0}^{b_1-1} C_i^{(2)} \mathbf{t}_i^{(1)}$.
-

In each round $r \in [k-1]$, the prover and verifier first construct the public parameters $\mathbf{pp}^{(k-r)} = (\mathbf{A}^{(k-r)}, \mathbf{B}^{(k-r)})$ based on the base commitment keys $\mathbf{A} \in \mathbb{R}_q^{\kappa \times b_0 \iota}$ and $\mathbf{B} \in \mathbb{R}_q^{\kappa \times \kappa \iota'}$ as follows:

$$\begin{aligned}\mathbf{A}^{(k-r)} &= \underbrace{\{1, 1, \dots, 1\}}_{\prod_{j=1}^{k-r-2} b_j \text{ ones}} \otimes \mathbf{A} = \vec{1}_{\prod_{j=1}^{k-r-2} b_j}^T \otimes \mathbf{A}, \\ \mathbf{B}^{(k-r)} &= \underbrace{\{1, 1, \dots, 1\}}_{b_{k-r-1} \text{ ones}} \otimes \mathbf{B} = \vec{1}_{b_{k-r-1}}^T \otimes \mathbf{B}.\end{aligned}$$

Next, the prover computes the folded vector

$$\text{fold}^{(k-r)} = \text{Fold}^{(k-r)} \left(\mathbf{s}^{(k-r)}, (\vec{a}_j)_{j \in [k-r-1]} \right)$$

according to Equation 4, and for each $i \in [b_{k-r-1}]$, computes the projected vector $\vec{\mathbf{p}}_i^{(k-r)} = \sigma_{-1}(\mathbf{\Pi}^{(k-r)}) D(\mathbf{s}_i^{(k-r-1)})$, where $\mathbf{\Pi}^{(k-r)} = \vec{1}_{\prod_{j=1}^{k-r-2} b_j}^T \otimes M_R(\Pi)$.

The prover then sends the proof:

$$\pi_r = \left(\text{fold}^{(k-r)}, \left(\vec{\mathbf{p}}_i^{(k-r)}, \mathbf{t}_i^{(k-r-1)} \right)_{i \in [b_{k-r-1}]} \right)$$

to the verifier.

By verifying that the norm of the constant component of $\vec{\mathbf{p}}_i^{(k-r)}$ is less than $\sqrt{128}\beta^{(k-r-1)}$, the verifier can deduce, with high probability, that the witness vector is bounded by $\sqrt{128/30}\beta^{(k-r-1)}$. Additionally, the verifier computes the inner product $\langle \text{fold}^{(k-r)}, \vec{a}_{k-r-1} \rangle$ to verify the correctness of the inner product constraint, and checks the binding between inner and outer commitments.

The verifier checks the correctness of π_r by performing consistency checks between adjacent rounds. Specifically, at the end of round r , the verifier samples a challenge $\vec{C}^{(k-r)} \in \mathcal{C}^{b_{k-r-1}}$ and sends it to the prover. Both parties then derive the next statement and witness for relation $\mathcal{R}_{k-r-1}$ using $\vec{C}^{(k-r)}$: as for the witness w_{r+1} , the prover folds the b_{k-r-1} slices of $\mathbf{s}^{(k-r)}$ to generate the next witness hypercube $\mathbf{s}^{(k-r-1)}$, together with the corresponding flattened vectors $D(\mathbf{s}_i^{(k-r-2)})$ and the inner commitments $(\mathbf{cm}_{\text{in},i}^{(k-r-2)})$ for all $i \in [b_{k-r-2}]$.

While the next round's statement is:

$$x_{r+1} = \left(\mathbf{pp}^{(k-r-1)}, (\vec{a}_j)_{j \in [k-r-1]}, \mathbf{y}^{(k-r-1)}, \mathbf{cm}_{\text{out}}^{(k-r-1)} \right),$$

where

$$\begin{aligned}\mathbf{pp}^{(k-r-1)} &= (\mathbf{A}^{(k-r-1)}, \mathbf{B}^{(k-r-1)}), \quad \mathbf{y}^{(k-r-1)} = \langle \text{fold}^{(k-r)}, \vec{C}^{(k-r)} \rangle, \\ \mathbf{cm}_{\text{out}}^{(k-r-1)} &= \mathbf{B} \left(G_{\delta_t, \kappa}^{-1} \left(\sum_{i=0}^{b_{k-r-1}-1} C_i^{(k-r)} \mathbf{cm}_{\text{in},i}^{(k-r-1)} \right) \right).\end{aligned}$$

And in round $r+1$, by checking the consistency condition:

$$\sum_{j=0}^{b_{k-r-1}-1} C_j^{(k-r)} \vec{\mathbf{p}}_j^{(k-r)} \stackrel{?}{=} \sum_{j=0}^{b_{k-r-2}-1} \vec{\mathbf{p}}_j^{(k-r-1)},$$

the verifier can deduce that the vectors $(\bar{\mathbf{p}}_i^{(k-r)})_{i \in [b_{k-r-1}]}$ sent in round r are consistent with the projections obtained in round $r+1$.

It is worth noting that the correctness of $\text{fold}^{(k-r)}$ and the inner commitments $(\text{cm}_{\text{in},i}^{(k-r-1)})_{i \in [b_{k-r-1}]}$ is implicitly verified through the same consistency check, as the updated value $\mathbf{y}^{(k-r-1)}$ and the outer commitment in round $r+1$ are both computed based on the proof provided in round r and the challenge $\vec{C}^{(k-r)}$.

After $k-1$ rounds, the prover generates a statement-witness pair in $\mathcal{R}_1$ and directly sends $\mathbf{s}^{(1)}$ to the verifier. The verifier then performs the final consistency checks:

$$\langle \sigma_{-1}(M_R(\vec{a}_0)), \mathbf{s}^{(1)} \rangle \stackrel{?}{=} \langle \text{fold}^{(2)}, \vec{C}^{(2)} \rangle$$

$$\sigma_{-1}(\Pi)\mathbf{s}^{(1)} \stackrel{?}{=} \sum_{i=0}^{b_1-1} C_i^{(2)} \bar{\mathbf{p}}_i^{(2)}, \quad \mathbf{A}\mathbf{s}^{(1)} \stackrel{?}{=} \sum_{i=0}^{b_1-1} C_i^{(2)} \mathbf{t}_i^{(1)}.$$

and the basic norm check on $\mathbf{s}^{(1)}$.

4.2 Security Analysis & Efficiency

Theorem 3. (Completeness) *Let the challenge space $\mathcal{C} \subset \mathbb{R}_q$ consist of ring elements with ℓ_2 norm bounded by τ , and operator norm bounded by T , where T is the operator norm defined in Section 2.1. Define $N = \left(\prod_{i=0}^{k-1} b_i\right) d$, $\iota = \lceil \log_\delta^q \rceil$, $\beta^{(k-1)} = \sqrt{\frac{N\iota}{b_{k-1}d}} \cdot \frac{\delta}{2}$ and recursively, $\beta^{(k-r-1)} = \frac{Tb_{k-r}}{\sqrt{b_{k-r-1}}} \cdot \beta^{(k-r)}$ for $r \in \{1, 2, \dots, k-1\}$. Then, Protocol 1 for proving relation $\mathcal{R}_k$ satisfies completeness.*

Proof. Apart from the initial step where the verifier sends Π to the prover, the remaining interaction can be viewed as a reduction of knowledge from $\mathcal{R}_k$ to $\mathcal{R}_1$, plus a proof for $\mathcal{R}_1$.

We establish the completeness of the entire protocol by show that the whole reduction process is complete. Specifically, for each $r \in [k]$, we prove the following:

1. Given that $(x_r, w_r) \in \mathcal{R}_{k-r}$, the tuple (x_{r+1}, w_{r+1}) produced in the r -th round satisfies $(x_{r+1}, w_{r+1}) \in \mathcal{R}_{k-r-1}$.
2. The verifier $\mathcal{V}$ accepts whenever $(x_r, w_r) \in \mathcal{R}_{k-r}$;

For each round $r \in [k]$, the statement-witness pair corresponding to the relation $\mathcal{R}_{k-r}$ is given by:

$$x_r = (\mathbf{A}^{(k-r)}, \mathbf{B}^{(k-r)}, (\vec{a}_j)_{j \in [k-r]}, \mathbf{y}^{(k-r)}, \text{cm}_{\text{out}}^{(k-r)}),$$

$$w_r = (\mathbf{s}^{(k-r)}, (D(\mathbf{s}_i^{(k-r-1)}), \text{cm}_{\text{in},i}^{(k-r-1)})_{i \in [b_{k-r-1}]}).$$

We first focus on the preservation of completeness across recursive reductions. At the end of round r , the prover computes the updated witness hypercube:

$$\mathbf{s}^{(k-r-1)} = \sum_{i=0}^{b_{k-r-1}-1} C_i^{(k-r)} \mathbf{s}_i^{(k-r-1)},$$

thus we have:

$$\begin{aligned}
& \left\langle \text{Fold}^{(k-r-1)} \left(\mathbf{s}^{(k-r-1)}, (\vec{a}_j)_{j \in [k-r-2]} \right), \vec{a}_{k-r-2} \right\rangle \\
&= \left(\cdots \left(\mathcal{F} \left(\mathcal{F} \left(\sum_{i=0}^{b_{k-r-1}-1} C_i^{(k-r)} \mathbf{s}_i^{(k-r-1)} \right) \cdot \sigma_{-1}(M_R(\vec{a}_0)) \right) \cdot \vec{a}_1 \right) \cdots \right) \cdot \vec{a}_{k-r-2} \\
&= \left\langle \text{Fold}^{(k-r)} \left(\mathbf{s}^{(k-r)}, (\vec{a}_j)_{j \in [k-r-1]} \right), \vec{C}^{(k-r)} \right\rangle \\
&= \left\langle \text{fold}^{(k-r)}, \vec{C}^{(k-r)} \right\rangle = \mathbf{y}^{(k-r-1)},
\end{aligned}$$

which ensures the inner product relation for $\mathcal{R}_{k-r-1}$ holds.

As for the norm constraint, since the updated witness vectors are:

$$D(\mathbf{s}_i^{(k-r-2)}) = \left(\sum_{j=0}^{b_{k-r-1}-1} C_j^{(k-r)} D(\mathbf{s}_j^{(k-r-1)}) \right) [il : (i+1)l - 1].$$

for $l = \text{len}(D(\mathbf{s}_i^{(k-r-2)})) = (\prod_{j=0}^{k-r-3} b_j)l$, we have

$$\|D(\mathbf{s}_i^{(k-r-2)})\| \leq \frac{T\beta^{(k-r-1)}}{\sqrt{b_{k-r-2}}} \cdot b_{k-r-1} = \beta^{(k-r-2)}.$$

As for the commitment binding relation, the inner commitment is computed from $D(\mathbf{s}_i^{(k-r-2)})$, so an honest prover will always pass the check.

For the outer commitment, the updated $\text{cm}_{\text{out}}^{(k-r-1)}$ computed by both parties is:

$$\text{cm}_{\text{out}}^{(k-r-1)} = \mathbf{B} \left(G^{-1} \left(\sum_{i=0}^{b_{k-r-1}-1} C_i^{(k-r)} \text{cm}_{\text{in},i}^{(k-r-1)} \right) \right).$$

Since

$$\begin{aligned}
& \text{cm}_{\text{in},i}^{(k-r-1)} \\
&= \mathbf{A}^{(k-r)} D(\mathbf{s}_i^{(k-r-1)}) \\
&= \underbrace{[\mathbf{A}^{(k-r-1)} \quad \mathbf{A}^{(k-r-1)} \quad \cdots \quad \mathbf{A}^{(k-r-1)}]}_{b_{k-r-2} \text{ times}} \cdot \begin{bmatrix} D(\mathbf{s}_i^{(k-r-1)})[0 : l - 1] \\ D(\mathbf{s}_i^{(k-r-1)})[l : 2l - 1] \\ \vdots \\ D(\mathbf{s}_i^{(k-r-1)})[(b_{k-r-2} - 1)l : b_{k-r-2}l - 1] \end{bmatrix} \\
&= \sum_{j=0}^{b_{k-r-2}-1} \mathbf{A}^{(k-r-1)} \left(D(\mathbf{s}_i^{(k-r-1)})[jl : (j+1)l - 1] \right),
\end{aligned}$$

thus,

$$\begin{aligned}
& \sum_{i=0}^{b_{k-r-1}-1} C_i^{(k-r)} \text{cm}_{\text{in},i}^{(k-r-1)} \\
&= \mathbf{A}^{(k-r-1)} \left(\sum_{i=0}^{b_{k-r-1}-1} C_i^{(k-r)} \sum_{j=0}^{b_{k-r-2}-1} D(\mathbf{s}_i^{(k-r-1)})[jl : (j+1)l-1] \right) \\
&= \mathbf{A}^{(k-r-1)} \left(\sum_{j=0}^{b_{k-r-2}-1} \sum_{i=0}^{b_{k-r-1}-1} C_i^{(k-r)} D(\mathbf{s}_i^{(k-r-1)})[jl : (j+1)l-1] \right) \\
&= \sum_{j=0}^{b_{k-r-2}-1} \text{cm}_{\text{in},j}^{(k-r-2)},
\end{aligned} \tag{5}$$

and,

$$\begin{aligned}
\text{cm}_{\text{out}}^{(k-r-1)} &= \mathbf{B}^{(k-r-1)} G_{\sigma_t, b_{k-r-2}\kappa}^{-1} \left(\left(\text{cm}_{\text{in},i}^{(k-r-1)} \right)_{i \in [b_{k-r-2}]} \right) \\
&= \underbrace{[\mathbf{B} \ \mathbf{B} \ \dots \ \mathbf{B}]}_{b_{k-r-2} \text{ times}} \begin{bmatrix} G_{\delta_t, \kappa}^{-1}(\text{cm}_{\text{in},0}^{(k-r-2)}) \\ G_{\delta_t, \kappa}^{-1}(\text{cm}_{\text{in},1}^{(k-r-2)}) \\ \vdots \\ G_{\delta_t, \kappa}^{-1}(\text{cm}_{\text{in}, b_{k-r-2}-1}^{(k-r-2)}) \end{bmatrix} \\
&= \sum_{j=0}^{b_{k-r-2}-1} \mathbf{B} \left(G_{\delta_t, \kappa}^{-1}(\text{cm}_{\text{in},j}^{(k-r-2)}) \right) \\
&= \mathbf{B} \left(G_{\delta_t, \kappa}^{-1} \left(\sum_{i=0}^{b_{k-r-1}-1} C_i^{(k-r)} \text{cm}_{\text{in},i}^{(k-r-1)} \right) \right).
\end{aligned}$$

Thus, when $(x_r, w_r) \in \mathcal{R}_{k-r}$, it follows that $(x_{r+1}, w_{r+1}) \in \mathcal{R}_{k-r-1}$.

Now we show that the verifier always accepts in round $r \in [k]$ when a pair of statement-witness is in $\mathcal{R}_{k-r}$. In the final round, the prover sends the witness directly to the verifier. By performing consistency and norm checks, the verifier will always accept if $w_{k-1} = \mathbf{s}^{(1)} \in \mathcal{R}_1(x_{k-1})$.

For other rounds, if $(x_r, w_r) \in \mathcal{R}_{k-r}$, the witness w_r satisfies the following:

1. The final inner product constraint involving $\text{fold}^{(k-r)}$ and $\vec{a}_{k-r-1}$;
2. The norm bound constraint on the flattened witness vectors $D(\mathbf{s}_i^{(k-r-1)})$;
3. Binding with the inner and outer commitments.

These conditions collectively ensure that the instance (x_r, w_r) passes the verifier's checks 1–3 in the r -th round.

As for the consistency checks on projection vectors (check 4 in each round), the verifier will accept as long as the proof from the previous round is correctly

computed, which is guaranteed by an honest prover because we have:

$$\begin{aligned}
& \sum_{i=0}^{b_{k-r-2}-1} \vec{\mathbf{p}}_i^{(k-r-1)} \\
&= \sigma_{-1}(\mathbf{\Pi})^{(k-r-1)} D(\mathbf{s}_i^{(k-r-2)}) \\
&= \sum_{i=0}^{b_{k-r-2}-1} \sigma_{-1}(\mathbf{\Pi})^{(k-r-1)} \left(\sum_{j=0}^{b_{k-r-1}-1} C_j^{(k-r)} D(\mathbf{s}_j^{(k-r-1)}) \right) [il : (i+1)l-1] \\
&= \sum_{j=0}^{b_{k-r-1}-1} C_j^{(k-r)} \vec{\mathbf{p}}_j^{(k-r)}.
\end{aligned}$$

Thus, when $(x_r, w_r) \in \mathcal{R}_{k-r}$, all of the verifier's checks in round r will pass. We therefore conclude that if the initial statement-witness pair $(x_0, w_0) \in \mathcal{R}_k$, then the verifier accepts in all rounds from 0 to $k-2$, and the final output of the recursive process, (x_{k-1}, w_{k-1}) , lies in $\mathcal{R}_1$, which is verified and accepted in the last round. Hence, the completeness of Protocol 1 is established. $\square$

Theorem 4. (Round-by-Round Coordinate-Wise Special Soundness) *Let*

$b_0 = b_1 = \dots = b_{k-1} = b = (\frac{N}{d})^{1/k}$, and set $\beta^{(k-1)} = \sqrt{\frac{N_t}{bd}} \cdot \frac{\delta}{2}$, $\beta^{(k-r-1)} = \sqrt{bT} \cdot \beta^{(k-r)}$ for $r \in \{1, 2, \dots, k-1\}$, with the constraint that $\beta^{(r)} \leq \sqrt{\frac{30}{128}} \cdot \frac{q}{125}$ for all $r \in [k]$. Additionally, define

$$\gamma = \max_{r \in [k]} (\sqrt{b} \beta^{(k-r)}), \quad \beta' = \sqrt{b\kappa\iota'} \cdot \frac{\delta_t}{2}, \quad \text{where } \iota' = \lceil \log_{\delta_t}^q \rceil.$$

Assuming that the MSIS problem is hard for rank κ and norm bound $\max(8T\gamma, \beta')$, Protocol 1 satisfies round-by-round coordinate-wise special soundness. Specifically, for each round $r \in [k-1]$, there exists a polynomial-time extractor Ext_r which, given $(b+1)$ accepting transcripts, can extract a valid witness $w_r \in \mathcal{R}_{k-r}(x_r)$ with soundness error at most 2^{-126} and a constant norm slack of $\sqrt{128/30} \approx 2$.

Proof. For $r \in [k-1]$, the extractor Ext_r takes $(b+1)$ accepting transcripts that includes $(x_r, \pi_r, c_{r,m}, w_{r+1,m})$ for $m \in [b+1]$, where

$$\begin{aligned}
x_r &= (\mathbf{A}^{(k-r)}, \mathbf{B}^{(k-r)}, (\vec{a}_j)_{j \in [k-r]}, \mathbf{Y}^{(k-r)}, \mathbf{cm}_{\text{out}}^{(k-r)}), \\
\pi_r &= \left(\text{fold}^{(k-r)}, (\vec{\mathbf{p}}_i^{(k-r)}, \mathbf{cm}_{\text{in},i}^{(k-r-1)})_{i \in [b]} \right), \\
c_{r,m} &= (\vec{C}_m^{(k-r)})_{m \in [b+1]}, \\
w_{r+1,m} &= (\mathbf{s}_m^{(k-r-1)}, (D(\mathbf{s}_{i,m}^{(k-r-2)}), \mathbf{cm}_{\text{in},i,m}^{(k-r-2)})_{i \in [b]})_{m \in [b+1]}.
\end{aligned}$$

for $(\vec{C}_m^{(k-r)})_{m \in [b+1]} \in \text{SS}(\mathcal{C}, b, 2)$ and $(x_{r+1,m}, w_{r+1,m}) \in \mathcal{R}_{k-r-1}$, can output a witness w_r such that $(x_r, w_r) \in \mathcal{R}_{k-r}$. The proof is structured as follows:

Extracting the weak commitment opening. By concatenating all the $(D(\mathbf{s}_{i,m}^{(k-r-2)}))_{i \in [b]}$, we obtain a long vector of length $b^{k-r-1} \iota$, which we denote as $\vec{\mathbf{S}}_m^{(k-r-1)}$. And this vector satisfies:

$$\vec{\mathbf{S}}_m^{(k-r-1)} = \sum_{i=0}^{b-1} \vec{C}_{m,i}^{(k-r)} D(\mathbf{s}_i^{(k-r-1)}). \quad (6)$$

Fix $v \in [b]$, and let $\vec{C}_b^{(k-r)}$ denote the center challenge such that other challenge $\vec{C}_v^{(k-r)}$ differs from it only in the v -th coordinate. Then, the extractor can compute a weak opening as follows:

$$\bar{D}(\mathbf{s}_v^{(k-r-1)}) = \frac{\vec{\mathbf{S}}_b^{(k-r-1)} - \vec{\mathbf{S}}_v^{(k-r-1)}}{\bar{c}_v^{(k-r)}}, \quad (7)$$

where $\bar{c}_v^{(k-r)} = \vec{C}_{b,v}^{(k-r)} - \vec{C}_{v,v}^{(k-r)}$, which satisfies the bound $\|\bar{c}_v^{(k-r)} \cdot \bar{D}(\mathbf{s}_v^{(k-r-1)})\| \leq 2\sqrt{b}\beta^{(k-r-2)}$.

By concatenating all $(\bar{D}(\mathbf{s}_i^{(k-r-1)}))_{i \in [b]}$ and reshaping into a $k-r$ dimensional hypercube, we obtain $\bar{\mathbf{s}}^{(k-r)}$. Since $(\mathbf{t}_i^{(k-r-1)})_{i \in [b]}$ is directly sent by the prover, the extractor can output it without modification.

Prove the binding constraint of the extracted witness. We now prove that the extracted witness satisfies the binding constraints.

As for the inner commitment binding constraint, we first show that for each $v \in [b]$, the relation $\mathbf{A}^{(k-r)} \bar{D}(\mathbf{s}_v^{(k-r-1)}) = \mathbf{cm}_{\text{in},v}^{(k-r-1)}$ holds, and then prove that no alternative weak opening can satisfy the same relation.

From Equation 5, we have:

$$\sum_{i=0}^{b-1} C_{m,i}^{(k-r)} \mathbf{cm}_{\text{in},i}^{(k-r-1)} = \sum_{i=0}^{b-1} \mathbf{cm}_{\text{in},i,m}^{(k-r-2)} = \mathbf{A}^{(k-r-1)} \sum_{i=0}^{b-1} \mathbf{s}_{i,m}^{(k-r-2)} = \mathbf{A}^{(k-r)} \vec{\mathbf{S}}_m^{(k-r-1)}.$$

Since $(x_{r+1,m}, w_{r+1,m}) \in \mathcal{R}_{k-r-1}$, the middle binding relation between $\mathbf{t}_{i,m}^{(k-r-2)}$ and $\mathbf{s}_{i,m}^{(k-r-2)}$ holds. This implies:

$$\sum_{i=0}^{b-1} C_{m,i}^{(k-r)} \mathbf{cm}_{\text{in},i}^{(k-r-1)} = \mathbf{A}^{(k-r)} \vec{\mathbf{S}}_m^{(k-r-1)}.$$

Combining Equations 6 and 7, we obtain $\bar{c}_v^{(k-r)} \mathbf{A}^{(k-r)} \bar{D}(\mathbf{s}_v^{(k-r-1)}) = \bar{c}_v^{(k-r)} \mathbf{cm}_{\text{in},v}^{(k-r-1)}$ and thus, $\mathbf{A}^{(k-r)} \bar{D}(\mathbf{s}_v^{(k-r-1)}) = \mathbf{cm}_{\text{in},v}^{(k-r-1)}$.

Now assume there exists an alternative weak opening $(\bar{D}^*(\mathbf{s}_v^{(k-r-1)}), (\bar{c}_v^{(k-r)})^*)_{v \in [b]}$ such that for all $v \in [b]$,

$$\|(\bar{c}_v^{(k-r)})^* \cdot \bar{D}^*(\mathbf{s}_v^{(k-r-1)})\| \leq 2\sqrt{b}\beta^{(k-r-2)}, \quad \mathbf{A}^{(k-r)} \bar{D}^*(\mathbf{s}_v^{(k-r-1)}) = \mathbf{cm}_{\text{in},v}^{(k-r-1)}.$$

Then for any $v \in [b]$, we have:

$$\begin{aligned}\bar{c}_v^{(k-r)} \mathbf{A}^{(k-r)} ((\bar{c}_v^{(k-r)})^* \bar{D}^*(\mathbf{s}_v^{(k-r-1)})) &= (\bar{c}_v^{(k-r)})^* \mathbf{A}^{(k-r)} (\bar{c}_v^{(k-r)} \bar{D}(\mathbf{s}_v^{(k-r-1)})) \\ &= \bar{c}_v^{(k-r)} (\bar{c}_v^{(k-r)})^* \mathbf{t}_v^{(k-r-1)},\end{aligned}$$

which implies:

$$\mathbf{A}^{(k-r)} \left(\bar{c}_v^{(k-r)} (\bar{c}_v^{(k-r)})^* \bar{D}^*(\mathbf{s}_v^{(k-r-1)}) - (\bar{c}_v^{(k-r)})^* \bar{c}_v^{(k-r)} \bar{D}(\mathbf{s}_v^{(k-r-1)}) \right) = \mathbf{0}.$$

Therefore, difference term on the left is the non-trivial solution of $\mathbf{A}^{(k-r)}$ and has norm at most $4T \cdot 2\sqrt{b}\beta^{(k-r-2)} = 8T\sqrt{b}\beta^{(k-r-2)}$. This directly contradicts the MSIS assumption for rank κ and norm bound $8T\gamma$.

As for the outer commitment binding constraint, the verifier computes the outer commitment using the prover-sent $(\mathbf{cm}_{\text{in},i}^{(k-r-1)})_{i \in [b]}$ and compares it with the statement's $\mathbf{cm}_{\text{out}}^{(k-r)}$. A successful verification implies that the outer commitment is correctly bound to $(\mathbf{cm}_{\text{in},i}^{(k-r-1)})_{i \in [b]}$ under the MSIS assumption with rank κ and norm bound β' .

Prove the extracted witness has small norm. We then show that the extracted witness satisfies the norm bound with overwhelming probability.

If the vectors $(\bar{\mathbf{p}}_i^{(k-r)})_{i \in [b]}$ are honestly computed, then by Lemma 3, the probability that $\|\bar{D}(\mathbf{s}_i^{(k-r-1)})\| > \sqrt{128/30} \cdot \beta^{(k-r-1)}$ for all $i \in [b]$ while still passing the check is at most 2^{-128} .

If $(\bar{\mathbf{p}}_i^{(k-r)})_{i \in [b]}$ are not honestly computed—i.e., there exists $i \in [b]$ and $j \in [256]$ such that $\mathbf{p}_{i,j}^{(k-r)} \neq \sigma_{-1}(\mathbf{\Pi}^{(k-r)})_j D(\mathbf{s}_i^{(k-r-1)})$, then the probability of passing the consistency check $\sum_{j=0}^{b-1} C_{m,i}^{(k-r)} \bar{\mathbf{p}}_i^{(k-r)} \stackrel{?}{=} \sum_{j=0}^{b-1} \bar{\mathbf{p}}_{i,m}^{(k-r-1)}$ in round $r+1$ is at most $1/(q^{d/2}) \leq 2^{-128}$.

This is because, given $(D(\mathbf{s}_{i,m}^{(k-r-2)}))_{i \in [b]}$, the verifier can independently compute the correct value of $\bar{\mathbf{p}}_{i,m}^{(k-r-1)}$, and the right-hand side of the consistency equation becomes deterministic. Therefore, the probability that $C_{i,m}^{(k-r)} \mathbf{p}_{i,j}^{(k-r)} = \sum_{i=0}^{b-1} \mathbf{p}_{i,m,j}^{(k-r-1)}$ holds is at most $q^{-d/2}$.

It is important to note that the correctness of the proof component $(\bar{\mathbf{p}}_i^{(k-r)})$ provided in round r is actually verified in the subsequent round $r+1$. Specifically, in the final check (round $k-1$), the prover directly sends $\mathbf{s}^{(1)}$, and the verifier computes the corresponding true value $\bar{\mathbf{p}}^{(1)}$ using $\sigma_{-1}(\mathbf{\Pi})\mathbf{s}^{(1)}$. Based on this correctly derived $\bar{\mathbf{p}}^{(1)}$, the verifier can infer the correctness of $(\bar{\mathbf{p}}_i^{(2)})_{i \in [b]}$, and recursively continue this verification upward, thereby confirming the validity of each $(\bar{\mathbf{p}}_i^{(k-r)})$ across all rounds.

Hence, the probability that the extracted witness violates the norm bound in each round is at most $2^{-128} + q^{-d/2} \leq 2^{-127}$.

Prove that the extracted witness satisfies the folded inner product relation. For an honestly computed value $\text{fold}^{(k-r)}$, if the extracted witness does not satisfy the IP relation, the probability that it passes the check $\langle \text{fold}^{(k-r)}, \vec{a}_{k-r-1} \rangle \stackrel{?}{=} \mathbf{y}^{(k-r)}$ is at most $1/q$.

While the probability that $\text{fold}^{(k-r)}$ is not honestly computed yet still passes verification is bounded by $q^{-d/2}$ in each round. We adopt the same recursive validation strategy for $\text{fold}^{(k-r)}$ as used for $(\bar{\mathbf{p}}_i^{(k-r)})$. In the final check (round $k-1$), the verifier verifies the correctness of $\text{fold}^{(2)}$ by checking whether

$$\langle \mathbf{s}^{(1)}, \sigma_{-1}(M_R(\vec{a}_0)) \rangle = \langle \text{fold}^{(2)}, \vec{C}^{(2)} \rangle = \mathbf{y}^{(1)},$$

holds. This allows the verifier to confirm that $\text{fold}^{(2)}$ is honestly computed from $\mathbf{s}^{(2)}$ with overwhelming probability of $(1 - q^{-d/2})$. By recursively applying this verification process from round $k-1$ upwards, the correctness of each $\text{fold}^{(k-r)}$ can be ensured.

Hence, the probability that the extracted witness does not satisfy the folded inner product constraint while pass the check in each round is at most $2^{-128} + q^{-d/2} \leq 2^{-127}$.

Therefore, in each round, the extractor Ext_r can successfully extract a valid witness

$$w_r = \left(\bar{\mathbf{s}}^{(k-r)}, \left(\bar{D}(\mathbf{s}_i^{(k-r-1)}) \right)_{i \in [b]}, \left(\mathbf{cm}_{\text{in}, i}^{(k-r-1)} \right)_{i \in [b]} \right)$$

in polynomial time, such that $w_r \in \mathcal{R}_{k-r}(x_r)$, with success probability at least $1 - 2^{-126}$ and a constant norm slack of $\sqrt{128/30}$. $\square$

Theorem 5. (Knowledge Soundness) *Protocol 1 for proving $\mathcal{R}_k$ is knowledge sound with total knowledge soundness error bounded by $\frac{(k-1)b}{|\mathcal{C}|} + 2^{-126}(k-1)$.*

Proof. According to Lemma 2 and Theorem 4, Protocol 1 satisfies round-by-round b -coordinate-wise 2-special soundness, with a knowledge soundness error of 2^{-126} per round. Applying the union bound over all $(k-1)$ rounds, the total soundness error is bounded by

$$\frac{(k-1)b}{|\mathcal{C}|} + 2^{-126}(k-1).$$

$\square$

Theorem 6. (Efficiency) *Let $b_0 = b_1 = \dots = b_{k-1} = b = (\frac{N}{d})^{1/k}$. Then, Protocol 1 constitutes an efficient proof system for the relation $\mathcal{R}_k$, achieving:*

- proof size of $O(kN^{1/k})$,
- verifier time of $O(kN^{1/k})$, and
- a linear prover cost, making it prover-friendly.

In particular, when $k = \log N - \log d$, the proof system attains $O(\log N)$ proof size and verification time.

Proof. In Protocol 1, at each round $r \in [k-1]$, the prover sends

$$\pi_r = \left(\text{fold}^{(k-r)}, \left(\vec{\mathbf{p}}_i^{(k-r)}, \text{cm}_{\text{in},i}^{(k-r-1)} \right)_{i \in [b]} \right),$$

where $\text{fold}^{(k-r)} \in \mathbb{R}_q^b$, each $\vec{\mathbf{p}}_i^{(k-r)} \in \mathbb{R}_q^{256}$, and $\text{cm}_{\text{in},i}^{(k-r-1)} \in \mathbb{R}_q^\kappa$. At the final round ($r = k$), the prover sends $\mathbf{s}^{(1)} \in \mathbb{R}_q^b$.

The total proof size is

$$((k-1)(257 + \kappa)b + b) \cdot \lceil \log q \rceil \text{ bits} = O(kN^{1/k}),$$

On the prover side, at each round, the dominant cost includes:

- Projection: $O(256b^{k-r}\iota)$ ring operations for computing $\vec{\mathbf{p}}_i^{(k-r)}$;
- Folding: $O(b^{k-r+1})$ integer operations for $\text{fold}^{(k-r)}$;
- Next-round witness and commitments: $O(b^{k-r}\iota + \kappa b^{k-r-1}\iota)$ ring operations;
- Statement update: $O(b + \kappa b + \kappa^2\iota')$ ring operations.

Thus, the total computational cost for each round is $O(b^{k-r})$. Since the prover performs these computations for $k-1$ rounds (i.e., $r \in [k]$), the total cost becomes:

$$O(b^k + b^{k-1} + \dots + b) = O(2b^k) = O(N)$$

Hence, the prover's cost is linear in the input size.

The verifier performs the following checks per round, totaling $O(N^{1/k})$ per round and $O(kN^{1/k})$ overall:

- Inner product: $O(b)$ ring operations;
- Norm checks on $\left(\vec{\mathbf{p}}_i^{(k-r)} \right)_{i \in [b]}$: $O(256b)$ ring operations;
- Consistency check on $\left(\vec{\mathbf{p}}_i^{(k-r)} \right)_{i \in [b]}$: $O(256b)$ ring operations;
- Commitment binding: $O(\kappa^2 b \iota')$ ring operations;
- Statement update: $O(b + \kappa b + \kappa^2 \iota')$.

□

5 Polynomial Commitment Scheme from HyperWolf

5.1 Basic Construction

In this section, we instantiate concrete parameters and employ the proof system described in Protocol 1 to construct a complete polynomial commitment scheme over $\mathbb{Z}_q$, suitable for both univariate and multilinear polynomials.

In the setup phase, given a security parameter λ , the algorithm outputs the public parameters $\text{pp} = (\mathbf{A} \in \mathbb{R}_q^{\kappa \times b_o \iota}, \mathbf{B} \in \mathbb{R}_q^{\kappa \times \kappa \iota'})$.

Let $\vec{f} = (f_0, f_1, \dots, f_{N-1})$, where each entry corresponds to the coefficient of either a univariate polynomial $f(X) := \sum_{i=0}^{N-1} f_i X^i$ or a multilinear polynomial

$f(X_0, X_1, \dots, X_{\ell-1}) := f_0 + f_1 X_0 + f_2 X_1 + \dots + f_{N-1} X_0 X_1 \dots X_{\ell-1}$, with $N = 2^\ell = d \cdot b^k$.

The commitment and opening procedures are identical for both cases and are described in Protocol 2. By instantiating concrete parameters and invoking Protocol 1, we obtain an efficient polynomial evaluation scheme applicable to both univariate and multilinear polynomials, as described in Protocol 3.

We also present batching techniques for handling more complex tasks, as explained in Appendix B.

Protocol 2 PC.Commit & PC.Open

Public parameters: $\text{pp} = (\mathbf{A} \in \mathbb{R}_q^{\kappa \times b_{ot}}, \mathbf{B} \in \mathbb{R}_q^{\kappa \times \kappa \ell'})$.

Commit ($\text{pp}, f \in \mathbb{Z}_q^{<N}[X]$ or $\mathbb{Z}_q[X_0, X_1, \dots, X_{\ell-1}]$) $\rightarrow$ (Com, st):

1. Represent f as $\vec{f} = (f_0, f_1, \dots, f_{N-1})$.
2. For $i = 0, 1, \dots, b^k - 1$, compute $\mathbf{f}_i = \sum_{j=0}^{d-1} f_{id+j} X^j \in \mathbb{R}_q$.
3. Let $\vec{\mathbf{f}} = (\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{b^k-1})$, and compute $\vec{\mathbf{s}} = G_{\delta, b^k}^{-1}(\vec{\mathbf{f}})$.
4. Parse $\vec{\mathbf{s}}$ into a k -dimensional hypercube $\mathbf{s}^{(k)}$ of shape $b \times \dots \times b \times b\ell$.
 - Let $(\mathbf{s}_i^{(k-1)})_{i \in [b]}$ be the slices of $\mathbf{s}^{(k)}$ along the k -th dimension.
 - Define $D(\mathbf{s}_i^{(k-1)}) \in \mathbb{R}_q^{b^{k-1}\ell}$ as the flattened vector of $\mathbf{s}_i^{(k-1)}$.
 - Let $\mathbf{A}^{(k)} = \vec{\mathbf{1}}_{b^{k-2}}^T \otimes \mathbf{A} \in \mathbb{R}_q^{\kappa \times b^{k-1}\ell}$.
 - For each $i \in [b]$, compute $\text{cm}_{\text{in}, i} = \mathbf{A}^{(k)} D(\mathbf{s}_i^{(k-1)})$.
5. Define $\text{cm}' = G_{\delta_t, b\kappa}^{-1}((\text{cm}_{\text{in}, i})_{i \in [b]})$ and $\text{cm}_{\text{out}} = \mathbf{B}^{(k)} \text{cm}'$, where $\mathbf{B}^{(k)} = \vec{\mathbf{1}}_b^T \otimes \mathbf{B} \in \mathbb{R}_q^{\kappa \times b\kappa\ell'}$.
6. Let $\text{st} := (\mathbf{s}^{(k)}, (D(\mathbf{s}_i^{(k-1)}), \text{cm}_{\text{in}, i})_{i \in [b]})$.
7. Return $(\text{cm}_{\text{out}}, \text{st})$.

Open($\text{pp}, \text{Com}, f, \text{st}, (c_i)_{i \in [b]}$) $\rightarrow \{0, 1\}$:

1. Represent f as $\vec{f} = (f_0, f_1, \dots, f_{N-1})$.
 2. For $i = 0, 1, \dots, b^k - 1$, compute $\mathbf{f}_i = \sum_{j=0}^{d-1} f_{id+j} X^j \in \mathbb{R}_q$.
 3. Let $\vec{\mathbf{f}} = (\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N/d-1})$ and compute $\vec{\mathbf{s}} = G_{\delta, b^k}^{-1}(\vec{\mathbf{f}})$.
 4. If $(D(\mathbf{s}_i^{(k-1)}))_{i \in [b]} \neq D(\mathbf{s}^{(k)}) \vee G_{\delta, b^k}(D(\mathbf{s}^{(k)})) \neq \vec{\mathbf{f}}$, return 0.
 5. For each $i \in [b]$:
 - If $\mathbf{A}^{(k)} D(\mathbf{s}_i^{(k-1)}) \neq \text{cm}_{\text{in}, i} \vee \mathbf{B}^{(k)} G_{\delta_t, b\kappa}^{-1}((\text{cm}_{\text{in}, i})_{i \in [b]}) \neq \text{cm}_{\text{out}}$, return 0.
 - If $\|c_i \cdot D(\mathbf{s}_i^{(k-1)})\| \geq \bar{\beta} \vee \|c_i\| \geq \bar{\tau} \vee c_i \notin \mathbb{R}_q^\times$, return 0.
 6. Return 1.
-

Protocol 3 PC.Eval

Public parameters: $\text{pp} = (\mathbf{A} \in \mathbb{R}_q^{\kappa \times b_o \iota}, \mathbf{B} \in \mathbb{R}_q^{\kappa \times \kappa \iota'})$.

$\text{Eval}(\text{pp}, \text{cm}, u \text{ (or } \vec{u} \in \mathbb{Z}_q^\ell), y; (f, \text{st}))$:

The prover $\mathcal{P}$ and verifier $\mathcal{V}$ proceed as follows:

1. **If** $f(X) \in \mathbb{Z}_q^{<N}[X]$ (univariate case), then for $i = 1, 2, \dots, k-1$, compute:
 $\vec{a}_i = (1, u^{b^i}, u^{2b^i}, \dots, u^{(b-1)b^i})$, $\vec{a}_0 = (1, u, u^2, \dots, u^{bd-1})$.
 2. **If** $f(X_0, X_1, \dots, X_{\ell-1}) \in \mathbb{Z}_q[X_0, X_1, \dots, X_{\ell-1}]$ (multilinear case), then
compute: $\vec{a}_i = \bigotimes_{j=i \log b}^{(i+1) \log b - 1} (1, u_j)$, $\vec{a}_0 = \bigotimes_{j=0}^{\log b + \log d - 1} (1, u_j)$.
 3. Update $\vec{a}_0 \leftarrow G_{\delta, bd}^T \vec{a}_0$.
 4. The prover $\mathcal{P}$ computes: $\mathbf{y} = \left\langle \text{Fold}^{(k)}(\text{st.s}^{(k)}, (\vec{a}_j)_{j \in [k-1]}), \vec{a}_{k-1} \right\rangle$.
 5. $\mathcal{P}$ and $\mathcal{V}$ define the instance-witness pair:
 $x = (\text{pp}^{(k)}, \text{cm}_{\text{out}}, (\vec{a}_j)_{j \in [k]}, \mathbf{y})$, $w = \text{st}$.
 6. The prover and verifier execute Protocol 1 to prove $(x, w) \in \mathcal{R}_k$.
 7. The verifier $\mathcal{V}$ checks whether $\text{ct}(\mathbf{y}) \stackrel{?}{=} y$.
-

Theorem 7. *The polynomial commitment PC defined in Protocol 2 and Protocol 3 satisfies evaluation completeness, weak binding, and knowledge soundness under the Module-SIS assumption.*

Proof. Theorem 3 establishes evaluation completeness, Theorem 4 guarantees weak binding, and Theorem 5 ensures knowledge soundness. $\square$

5.2 Concrete Parameters

For $N = b^k d$ with $b = 2$ and $k = \log(N/d)$, the asymptotic performance of the polynomial commitment scheme PC defined in Protocols 2 and 3 in Table 3,

Commit. Size	Eval. Proof Size	Prover Cost	Verifier Cost	Soundness Error
$O(1)$	$O(\log N)$	$O(N)$	$O(\log N)$	$\frac{2(\log N - 7)}{ C } + 2^{-126}(\log N - 7)$

Table 3: Asymptotic performance

Table 4 gives the notation as well as the corresponding asymptotic parameters.

According to Theorem 4, the soundness holds when MSIS problem is hard for rank κ and norm bound $\max\left(8T\gamma, \sqrt{b\kappa\iota'} \cdot \frac{\delta_\iota}{2}\right)$, where $\gamma = \sqrt{2}\beta^{(0)} = \sqrt{2} \cdot (\sqrt{2}T)^{k-1} \cdot \sqrt{\frac{N_\iota}{2d}} \cdot \frac{\delta}{2}$, and $\beta^{(0)}$ should be less than $\sqrt{\frac{30}{128} \frac{q}{125}}$. As discussed in Section 2.2, we require that $\kappa \geq 2\iota$ and that the norm bound for MSIS assumption

Notation	Explanation	Asymptotic Instantiation
λ	Security parameter	128
q	Prime modulus	$O(\lambda)$, $q \equiv 5 \pmod{8}$
d	Ring dimension	Power of two, 64
N	Number of coefficients for f	$N = b^k d$
b	Length of auxiliary vectors	2
k	Dimension of the coefficient hypercube	$O(\log N)$
κ	Height of matrices $\mathbf{A}, \mathbf{B}$	$O(1)$
δ	Decomposition base 1	$q^{1/O(1)}$
ι	$\lceil \log_\delta q \rceil$	$O(1)$
δ_t	Decomposition base 2	$q^{1/O(1)}$
ι'	$\lceil \log_{\delta_t} q \rceil$	$O(1)$
τ	Norm of a challenge	$\omega(1)$
T	Operation norm bound of a challenge	$\omega(1)$
$\beta^{(k-r)}$	Norm bound of witness in round r	$< q/125$
$\mathcal{C}$	Challenge space	$ \mathcal{C} \approx 2^{-128}$

Table 4: Notations with asymptotic parameters

should be less than $\min\left(q, 2^{2\sqrt{d\kappa \log q \log(1.0045)}}\right)$. By choosing parameters $\delta = \delta_t = 16$, $\kappa = 2\iota$, and $T = 15$, we conclude that the number of coefficients N should satisfy $N < 2^{32}$.

As for the soundness error, by setting the ring as $\mathbb{R}_q[X] = \mathbb{Z}_q[X]/(X^{64} + 1)$, and the challenge to be the ring elements with 23 zero coefficients, 31 coefficients that are ± 1 , and 10 coefficients that are ± 2 (which is the same as the challenge set instantiated in **Labrador** [BS23]), we guarantee that the challenge space is more than 2^{128} . Furthermore, by adopting reject sampling 5-6 times, we can restrict the challenge operation norm to be less than 15. In such a case, the soundness error can be bound to $O(\frac{1}{2^{\lambda}})$ which is negligible.

6 Future Work

In the future, we plan to implement our scheme in Rust and develop a lattice-based PCS library. Additionally, we will optimize the ring mapping procedure by using the NTT to replace the current method of only extracting the constant term, thereby improving data utilization. Moreover, we will enhance the batching methods by proposing a more efficient and detailed batching scheme.

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A Simple Protocol for the case $k = 3$

Protocol 4 presents a simple example for the case $k = 3$, where we set $N = b_2 b_1 b_0 d$. And we use $\vec{1}_m$ to denote the all-one vector of length m .

B Batching

In this section, we present a method for batching multiple polynomial evaluation proofs. Specifically, we provide solutions to handle the following three cases:

Multiple Polynomials Evaluated at a Single Point. When the prover needs to demonstrate that $f_i(u) = v_i$ (or $f_i(\vec{u}) = v_i$ in the multilinear case) for a collection of polynomials $(f_i)_{i \in [n]}$ at the same evaluation point, the verifier first samples a random challenge vector $\vec{\alpha} \in \mathbb{Z}_q^n$. The prover then computes a linear combination of the polynomials as $f = \sum_{i=0}^{n-1} \alpha_i f_i$, and both parties compute the corresponding weighted evaluation $y = \sum_{i=0}^{n-1} \alpha_i v_i$. Finally, they run the evaluation protocol on the combined polynomial f and the claimed evaluation y at point u (or $\vec{u}$).

Single Polynomial Evaluated at Multiple Points. Consider the scenario where the prover needs to prove that $f(u_i) = v_i$ for multiple evaluation points ($i \in [n]$). The construction differs slightly between the univariate and multilinear settings.

Multilinear Case: The prover first constructs the formal multilinear extension $\tilde{f}$ of f as

$$\tilde{f}(\vec{x}) = \sum_{\vec{b} \in \{0,1\}^{\log N}} f(\vec{b}) \cdot \tilde{\mathbf{eq}}(\vec{b}, \vec{x}),$$

where $\tilde{\mathbf{eq}}(\vec{b}, \vec{x}) := \prod_{i=0}^{\log N - 1} (b[i]x[i] + (1 - b[i])(1 - x[i]))$, which satisfies $\tilde{f}(\vec{x}) = f(\vec{x})$ for all $\vec{x} \in \mathbb{Z}_q^{\log N}$. Hence, $\tilde{f}(\vec{u}_i) = f(\vec{u}_i) = v_i$ holds.

Next, the prover constructs the polynomial

$$g(\vec{x}) = \sum_{i=0}^{n-1} \alpha_i \cdot \tilde{f}(\vec{x}) \cdot \tilde{\mathbf{eq}}(\vec{x}, \vec{u}_i),$$

where $\vec{\alpha} \in \mathbb{Z}_q^n$ is a challenge vector sampled by the verifier.

Then the prover and verifier engage in a sum-check protocol to verify the claim:

$$\sum_{i=0}^{n-1} \alpha_i v_i = \sum_{\vec{b} \in \{0,1\}^{\log N}} g(\vec{b}).$$

This reduces to checking whether $f(\vec{r}) = v$ and $\tilde{\mathbf{eq}}(\vec{r}, \vec{u}_i) = z_i$ for some random $\vec{r}$ determined by the verifier. The latter can be computed by the verifier, while the former can be verified via the polynomial commitment scheme proposed in Section 5.

Protocol 4 Example: Proof System for $\mathcal{R}_3$

Public inputs:

$\mathbf{A} \in \mathbb{R}_q^{\kappa \times b_0 \iota}$, $\mathbf{B} \in \mathbb{R}_q^{\kappa \times \kappa \iota'}$, $\vec{a}_2 \in \mathbb{Z}_q^{b_2}$, $\vec{a}_1 \in \mathbb{Z}_q^{b_1}$, $\vec{a}_0 \in \mathbb{Z}_q^{b_0 \iota}$, $\mathbf{y}^{(3)} \in \mathbb{Z}_q$, $\mathbf{cm}_{\text{out}}^{(3)}$

Private inputs:

$\mathbf{s}^{(3)} \in \mathbb{R}^{b_2 \times b_1 \times b_0 \iota}$, $(D(\mathbf{s}_i^{(2)})) \in \mathbb{R}_q^{b_1 b_0 \iota}$, $\mathbf{cm}_{\text{in},i}^{(2)} \in \mathbb{R}_q^\kappa)_{i \in [b_2]}$

Round 0: $\mathcal{R}_3 \rightarrow \mathcal{R}_2$

1. $\mathcal{V} \rightarrow \mathcal{P}$: $\Pi \leftarrow \mathcal{D}^{256 \times b_0 \iota d}$.
2. $\mathcal{P}$ and $\mathcal{V}$ construct: $\mathbf{A}^{(3)} = \vec{1}_{b_1}^T \otimes \mathbf{A}$, $\mathbf{B}^{(3)} = \vec{1}_{b_2}^T \otimes \mathbf{B}$, $\mathbf{\Pi}^{(3)} = \vec{1}_{b_1}^T \otimes M_R(\Pi)$.
3. $\mathcal{P}$ computes: $\text{Fold}^{(3)}$ and $\vec{\mathbf{p}}_i^{(3)} = \sigma_{-1}(\mathbf{\Pi}^{(3)})D(\mathbf{s}_i^{(2)})$ for all $i \in [b_2]$.
4. $\mathcal{P} \rightarrow \mathcal{V}$: $\text{Fold}^{(3)}$, $(\vec{\mathbf{p}}_i^{(3)}, \mathbf{cm}_{\text{in},i}^{(2)})_{i \in [b_2]}$.
5. $\mathcal{V}$ checks:
 - (a) $\langle \text{Fold}^{(3)}, \vec{a}_2 \rangle \stackrel{?}{=} \mathbf{y}^{(3)}$;
 - (b) $\sum_{j=0}^{255} \left(\text{ct}(\vec{\mathbf{p}}_{i,j}^{(3)}) \right)^2 \leq 128 \cdot (\beta^{(2)})^2$, for all $i \in [b_2]$;
 - (c) $\mathbf{B}^{(3)}(G_{\delta_t, b_2 \kappa}^{-1}((\mathbf{cm}_{\text{in},i}^{(2)})_{i \in [b_2]})) \stackrel{?}{=} \mathbf{cm}_{\text{out}}^{(3)}$.
6. $\mathcal{V} \rightarrow \mathcal{P}$: $\vec{C}^{(3)} \in \mathcal{C}^{b_2}$.
7. $\mathcal{P}$ computes:

$$\mathbf{s}^{(2)} = \sum_{i=0}^{b_2-1} C_i^{(3)} \mathbf{s}_i^{(2)} \in \mathbb{R}_q^{b_1 \times b_0 \iota}, \quad (D(\mathbf{s}_i^{(1)})) \in \mathbb{R}_q^{b_0 \iota}, \mathbf{cm}_{\text{in},i}^{(1)} \in \mathbb{R}_q^\kappa)_{i \in [b_1]}.$$
8. $\mathcal{P}$ and $\mathcal{V}$ construct: $\mathbf{A}^{(2)} = \mathbf{A}$, $\mathbf{B}^{(2)} = \vec{1}_{b_1}^T \otimes \mathbf{B}$, $\mathbf{\Pi}^{(2)} = \mathbf{\Pi} = M_R(\Pi)$.
9. $\mathcal{P}$ and $\mathcal{V}$ compute:

$$\mathbf{y}^{(2)} = \langle \text{Fold}^{(3)}, \vec{C}^{(3)} \rangle, \quad \mathbf{cm}_{\text{out}}^{(2)} = \mathbf{B}(G_{\delta_t, \kappa}^{-1}(\sum_{i=0}^{b_2-1} C_i^{(3)} \mathbf{cm}_{\text{in},i}^{(2)})).$$

Round 1: $\mathcal{R}_2 \rightarrow \mathcal{R}_1$

1. $\mathcal{P}$ computes: $\text{Fold}^{(2)} = \mathbf{s}^{(2)} \cdot \sigma_{-1}(M_R(\vec{a}_0))$, $\vec{\mathbf{p}}_i^{(2)} = \mathbf{\Pi} D(\mathbf{s}_i^{(1)})$ for $i \in [b_1]$.
2. $\mathcal{P} \rightarrow \mathcal{V}$: $\text{Fold}^{(2)}$, $(\vec{\mathbf{p}}_i^{(2)}, \mathbf{cm}_{\text{in},i}^{(1)})_{i \in [b_1]}$.
3. $\mathcal{V}$ checks:
 - (a) $\langle \text{Fold}^{(2)}, \vec{a}_1 \rangle \stackrel{?}{=} \mathbf{y}^{(2)}$;
 - (b) $\sum_{j=0}^{255} \left(\text{ct}(\vec{\mathbf{p}}_{i,j}^{(2)}) \right)^2 \leq 128 \cdot (\beta^{(1)})^2$, for all $i \in [b_1]$;
 - (c) $\sum_{i=0}^{b_2-1} C_i^{(3)} \vec{\mathbf{p}}_i^{(3)} \stackrel{?}{=} \sum_{i=0}^{b_1-1} \vec{\mathbf{p}}_i^{(2)}$;
 - (d) $\mathbf{B}^{(2)}(G_{\delta_t, b_1 \kappa}^{-1}((\mathbf{cm}_{\text{in},i}^{(1)})_{i \in [b_1]})) \stackrel{?}{=} \mathbf{cm}_{\text{out}}^{(2)}$.
4. $\mathcal{V} \rightarrow \mathcal{P}$: $\vec{C}^{(2)} \in \mathcal{C}^{b_1}$.
5. $\mathcal{P}$ computes: $\mathbf{s}^{(1)} = \sum_{i=0}^{b_1-1} C_i^{(2)} \mathbf{s}_i^{(1)} \in \mathbb{R}_q^{b_0 \iota}$.
6. $\mathcal{P}$ and $\mathcal{V}$ compute:

$$\mathbf{y}^{(1)} = \langle \text{Fold}^{(2)}, \vec{C}^{(2)} \rangle, \quad \mathbf{cm}_{\text{out}}^{(1)} = \mathbf{B}(G_{\delta_t, \kappa}^{-1}(\sum_{i=0}^{b_1-1} C_i^{(2)} \mathbf{cm}_{\text{in},i}^{(1)})).$$

Round 2:

1. $\mathcal{P} \rightarrow \mathcal{V}$: $\mathbf{s}^{(1)}$.
 2. $\mathcal{V}$ checks:
 - (a) $\langle \mathbf{s}^{(1)}, \sigma_{-1}(M_R(\vec{a}_0)) \rangle \stackrel{?}{=} \mathbf{y}^{(1)}$; $\|\mathbf{s}^{(1)}\| \leq \beta^{(0)}$
 - (b) $\sigma_{-1}(\mathbf{\Pi})\mathbf{s}^{(1)} \stackrel{?}{=} \sum_{i=0}^{b_1-1} C_i^{(2)} \vec{\mathbf{p}}_i^{(2)}$; $\mathbf{A}\mathbf{s}^{(1)} \stackrel{?}{=} \sum_{i=0}^{b_1-1} C_i^{(2)} \mathbf{cm}_{\text{in},i}^{(1)}$.
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Univariate Case: Observe that if we define $X_i := X^{2^i}$, then the univariate polynomial

$$f(X) = f_0 + f_1X + f_2X^2 + \cdots + f_{N-1}X^{N-1}$$

can be rewritten in a multilinear form as

$$f(X_0, X_1, \dots, X_{\ell-1}) = f_0 + f_1X_0 + f_2X_1 + \cdots + f_{N-1}X_0X_1 \cdots X_{\ell-1},$$

with the same coefficient vector $\vec{f} = (f_0, f_1, \dots, f_{N-1})$. By constructing $\vec{u}_i = (u_i, u_i^2, \dots, u_i^{2^{\ell-1}})$, we obtain that $f(\vec{u}_i) = f(u_i)$.

Thus, the protocol reduces to the multilinear setting, with $\vec{u}_i$ in a special structured form, and the remaining steps proceed as in the multilinear case, ultimately reducing to a single-point evaluation check at a random point $\vec{r}$.

Multiple Polynomials Evaluated at Multiple Points. This case addresses the scenario where the prover must prove $f_i(\vec{u}_i) = v_i$ for multiple polynomials $(f_i)_{i \in [n]}$ evaluated at distinct points $(\vec{u}_i)_{i \in [n]}$.

The construction extends naturally from the second case. The verifier first samples a random challenge vector $\vec{\alpha} \in \mathbb{Z}_q^n$, and the prover constructs the combined polynomial:

$$g(\vec{x}) = \sum_{i=0}^{n-1} \alpha_i \cdot f_i(\vec{x}) \cdot \text{eq}(\vec{x}, \vec{u}_i).$$

The prover and verifier then engage in a sum-check protocol to verify:

$$\sum_{i=0}^{n-1} \alpha_i v_i \stackrel{?}{=} \sum_{\vec{b} \in \{0,1\}^{\log N}} g(\vec{b}).$$

As in the previous case, the verification ultimately reduces to checking the evaluations of the individual polynomials f_i at a randomly sampled point $\vec{r}$, which can be handled using the batching technique described in the first case.